Schlumberger
Carbon Services

Risk Management Topical Report: Illinois Basin – Decatur Project

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17 December 2015

CS1512-116-NWM

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1. Executive Summary

This report summarizes the Risk Management processes applied to the Illinois Basin – Decatur Project (IBDP) and the subsequent results. The primary objectives for the Risk Management processes applied to the IBDP were to:

1) Identify and treat important risks to project success; and
2) Develop iterative risk management procedures in a general carbon capture and storage (CCS) context.

The IBDP is one of seven Regional Carbon Sequestration Partnerships initiated by the US Department of Energy to carry out regional characterization and field projects to demonstrate the capability to permanently and safely store carbon dioxide (CO_2) and provide the information basis for commercial-scale projects. This endeavor was carried out in three phases:

- Phase I: Characterization,
- Phase II: Validation, and
- Phase III: Development.

This report focuses on the Phase III Risk Assesment and Management efforts of the IBDP.

To date, the IBDP has conducted three workshops at which risks were identified and evaluated. The first and second workshop sessions were conducted in Stage 1 (at project start-up) and the third was conducted in Stage 2, after a substantial tonnage of CO_2 had been injected. All three workshops involved a broad spectrum of project staff and outside experts, and resulted in a prioritized (ranked) list of risks based on the numerically aggregated, expert-weighted opinion of all participants.

To develop detailed risk management plans and the potential treatments to be applied, the Stage 1 risk workshops were organized in breakout groups by discipline; this allowed a smaller number of participants to focus on developing specific risk response actions (risk treatments) for potential implementation. Technological advancements enabled the Stage 2 workshop to be conducted entirely in plenary session.

The risk-bearing scenarios and treatments identified in the Stage 1 workshops centered primarily on topics related to project design and construction and geologic uncertainties. These areas had been largely addressed by the beginning of the Stage 2 workshop, where the primary issues included the potential for felt seismicity and the challenges of maintaining consistent plume tracking, largely due to challenges resulting from new applications of technologies. Risk Management topic areas for all three workshops also focused on plans and treatments for risks associated with health and safety, budget, and public interactions.

In Stage 1, substantial effort was applied to establish a tracking system for the execution of risk treatments and the resultant effects of the treatments on risk levels. After an initial staff-intensive period of implementation, the system was streamlined to focus on two goals:

1) Providing adequate information to support anticipated project decision points; and
2) Maintaining a culture of risk awareness throughout the project's lifecycle.

Although Stage 1 data capture on risk treatment and tracking status proved to be challenging, the process led to important lessons learned, and the collected information was adequate to support the decision to initiate CO_2 injection in November 2011.

Additionally, outside the purview of this report, perhaps the most powerful risk management tool was the culture of risk awareness that was cultivated and facilitated in the field and during planning operations during which numerous safety meetings were held that included operations specific job safety analyses based upon the scenarios and events described within this report.

2. Abbreviations and Acronyms Used

AOR	Area of Review
CCS	Carbon Capture and Storage
CCS1	Injection Well #1 (Illinois Basin – Decatur Project)
CCS2	Injection Well #2 (Illinois Industrial Carbon Capture and Storage Project)
CCUS	Carbon Capture, Utilization, and Storage
DOE	Department of Energy
EPA	Environmental Protection Agency
GS	Geological Sequestration
HARC	Hazard Analysis and Risk Control
HAZOP	Hazardous Operations
HSE	Health, Safety, and Environment
IBDP	Illinois Basin – Decatur Project
ICCS	Industrial Carbon Capture and Storage
IDLH	Immediate Danger to Life or Health
IEPA	Illinois Environmental Protection Agency
ISGS	Illinois State Geological Survey
JMP	Journey Management Plan
MGSC	Midwest Geological Sequestration Consortium
MVA	monitoring, verification, and accounting
NGO	Non-governmental Organization
NSEC	National Sequestration Education Center
OWG	Outreach Working Group
QHSE	Quality, Health, Safety, and Environment
RACI	Responsible, Accountable, Consulted, Informed
ROCI	Responsible, Oversight, Consulted, Informed
RRAG	Risk Response Action Group
SIG	Special Interest Group
SOPO	Statement of Project Objectives
STP	Site Traffic Plan
UIC	Underground Injection Control
USDW	Underground Source of Drinking Water
VSP	Vertical Seismic Profile
VW1	Verification Well #1 (Illinois Basin – Decatur Project)
VW2	Verification Well #2 (Illinois Industrial Carbon Capture and Storage Project)

3. Risk Management Objectives

3.1 Scope and Purpose

This report discusses risk management activities beginning with initial project funding of the Illinois Basin – Decatur Project (IBDP) in December 2007 and progressing through the end of 2014 by:

- Defining the purpose and objectives of risk management activities for the IBDP;
- Detailing the activities through which the objectives have been addressed;
- Summarizing the different risk issues deemed important at stages from project start through mid-project; and
- Noting changes to procedures that occurred during the course of the IBDP.

The following risk management activities are specifically documented and discussed in this report:

- Stage 1 Risk Assessment Workshop (parts 1 and 2), February/July, 2008;
- Stage 1 Risk Management Workplan establishment and execution;
- Pre-injection risk-evaluation stage gate;
- Stage 2 Interim Risk Assessment Workshop, 2013; and
- Stage-2 Risk Management Workplan establishment and initial execution.

Although outside the purview of this report, perhaps the most powerful risk management tool was the culture of risk awareness that was cultivated and facilitated in the field and during planning operations during which numerous safety meetings were held that included operations specific job safety analyses based upon the scenarios and events described within this report.

Prior publications (Appendix 7.9) have addressed risk management techniques and outcomes for the IBDP at varying levels of detail. Noteworthy among these publications are papers written in connection with the Greenhouse Gas Technologies (GHGT) conference series, presentations published in connection with the International Energy Agency Greenhouse Gas Program (IEAGHG) Risk Assessment Network meetings, and the risk management chapter produced for a special volume published by Greenhouse Gases: Science and Technology (Wiley).

3.2 Basis for Risk Management

Statement of Project Objectives

The original Statement of Project Objectives (SOPO) from February, 2008, noted:

> *The goal of the Phase III amendment to the work of the Midwest Geological Sequestration Consortium (MGSC) is to demonstrate the ability of the Mt. Simon Sandstone, a major regional saline reservoir in the Illinois Basin, to accept and retain 1 million ton[ne]s of carbon dioxide (CO_2) injected over a period of three years.*

Although the 2008 SOPO has been updated periodically to reflect project progress, the mission of risk management has remained consistently aligned with project purpose. Both the original SOPO and its revisions identified task objectives in 14 areas, and risk management has been organized to make positive contributions toward execution in each area. Risk management is an integral and indispensable part of project management and provides a tool for addressing the many complexities of large-volume CO_2 injection demonstration projects such as the IBDP.

Summary of Risk Management Objectives

Although the 2008 and 2012 SOPOs do not specify objectives in risk management itself, risk management activities and information are fundamental to planning most other aspects of project design and execution. SOPO statements also establish the key concept in risk assessment called *Project Values* (see page 11). The 2008 and 2012 SOPOs imply objectives in risk management as follows:

- Thoroughly identify sources of risk to all aspects of project success.
- Characterize risk pathways and quantify (to the extent possible) risk magnitudes.
- Evaluate risk magnitudes and develop plans to reduce unacceptably high risks.
- Track the status of risk and risk treatment, so that adequate information is available at stage gates and decision points.
- Support IBDP success through risk management.
- Foresee and prevent risks to project objectives.
- Draw lessons from the IBDP experience to guide risk management activities in geological sequestration (GS) projects generally.
- Develop risk management methods that are applicable to and aligned with GS projects.

Global Context

Due to its unique status as a research project, the IBDP should be viewed not only in the context of its internal SOPO, but in the larger context of the global GS standards that have influenced it. International guidance or standards that have influenced risk management for the IBDP are cited in Table 1. Several individual participants in IBDP[1] risk management processes have also worked in the development of these standards.

Table 1. Standards and Guidance on GS Risk Management

Standard or Guidance	Note
EU Directive	In place at the beginning of the IBDP.
WRI CCS Guidelines[1]	In development during early stages of the IBDP. Issued in 2008.
Canadian Standards Association (CSA) CCS Standard z-741	In development during early stages of the IBDP. Issued in 2012.
International Standards Organization (ISO) Technical Committee 265[1]	Currently in development, ISO CCS Standards expected to be issued in 2016.

For all of the standards listed in Table 1 and future standards, the focus is on long-term, commercial-scale GS projects. The IBDP is a demonstration project with an injection period of three years and thus, is scaled differently than a commercial project. The IBDP contributed to the development of the standards. The IBDP was under development in 2007 and pre-dates many of the standards currently in place or being developed. The research at IBDP contributed to standards developed since 2008.

[1] S. Greenberg, K. Hnottavange-Telleen, J. Kirksey

4. Risk Management for the IBDP

4.1 Scope and Terminology

The scope of risk management for the IBDP corresponds directly with the scope of the IBDP itself. The aims of the project — expressed in the SOPO and the Project Values (page 11) — describe the entities against which risk is identified, analyzed, evaluated, and treated. Given this extensive scope, the broad purpose of risk management is to clear paths toward the project's goals by identifying and removing potential challenges (wherever possible), and to proactively continue doing so as the project progresses, and changes occur in its status, knowledge base, and environment.

Project risks are managed through the chained processes of risk assessment, risk evaluation, treatment definition, treatment tracking, and re-assessment, and evaluation. For the IBDP, risk assessment focused initially on broad entities called Features, Events, and Processes (FEPs), and later on more specific entities called scenarios. For the risk assessment performed at project start-up, fundamental risk concepts were defined (including project values and risk scales), and project information was reviewed to create a common basis for evaluation. Risk assessment results (based on FEPs) were applied to drive the creation of a set of scenarios, which in turn were applied to create a set of risk treatments. Tracking the completion status of the risk treatments continues to provide the basis for re-evaluating risk at critical project junctures and decision points such as the start of injection. These processes are detailed in Section 4.6, *Risk Treatment and Risk Tracking* (page 23).

Terminology related to risk management processes are not identical in all contexts, and continues to evolve even within the context of GS. The terminology used in this report (see Table 2) generally follows the definitions in International Standards Organization Risk Management Standard ISO 31000 and those in Canadian Standards Association CCS Standard z741-2012.

Table 2: Risk Terminology

Term	Definition or Description
FEP	Acronym for Features, Events, and Processes. FEPs are elements, concepts, components, or outside factors that are important to an activity (a project) for which risk is being evaluated. Because a FEP may participate in multiple scenarios, the risk attributes evaluated for a FEP represent an aggregate.
Likelihood	The likelihood that negative impact will occur *at a given level of severity* to one or more project values. This definition requires that, for any FEP or scenario whose risk is evaluated, severity must be evaluated first and likelihood second. Because the probabilities of event occurrence may range from very small to very large, likelihood is semi-quantified on a categorical 1-to-5 scale.
Project Values	The valued entities, conditions, and/or objectives identified by the risk owner as being potentially at risk. In other contexts, similar terms include elements of concern and risk receptors. Risk to project values may arise directly because of project activities, indirectly due to exposures related to project activities, or because of inadequate or incompetent performance of project activities.
Risk Analysis	The process of investigating the sources, pathways, and magnitudes (levels of severity and likelihood) of risk. Risk analysis outcomes may be described in quantitative or qualitative terms.
Risk Assessment	An umbrella term whose meaning includes risk identification, risk analysis, and risk evaluation. In some contexts, the meaning is broader and includes risk management.

Term	Definition or Description
Risk Attributes and Risk Quantification	The risk attributes used here are severity (S) and likelihood (L). S and L are expressed on 1-to-5 scales, but the number assignment is progressively categorical rather than quantitative. The product severity times likelihood (S^xL) is identified as the quantity of Risk, which therefore scales from 1 to 25.
Risk Evaluation	The process of comparing criteria to the results of risk analysis, to determine whether a specific risk instance: a) is acceptable or tolerable; vs. b) must be treated in order to reduce or maintain low risk.
Risk Evaluation Criteria	Standards against which the significance of risk is evaluated. The definition of these criteria represents judgment and policy rather than measurement, even if the criteria are expressed quantitatively. One fundamental risk evaluation criterion is the set of identified project values; another is the expression of the level of risk that is deemed acceptable.
Risk Identification	The process of finding, recognizing, and describing risks.
Risk Management	In this report, the term risk management is broadly applied. All processes listed below are viewed as components of risk management.
Risk Management Plan	A written description of the methods and resources to be applied to the management of risks.
Risk Owner	A person or entity that has the accountability, authority, and resources to manage risk.
Risk Treatment	An activity designed to reduce risk or to maintain risk at an acceptably low level.
Scenario or Risk Scenario	A chain of events that: a) may plausibly occur within the context being considered (e.g., the project), and b) may lead to negative impact upon one or more project values. Because an infinite number of scenarios might occur during a project, each specified scenario is understood as a proxy for various chains of events whose risk attributes and treatments are similar. A scenario is expressed in a complete sentence, and the semi-quantitative risk attributes of likelihood and severity are most discretely evaluated with respect to a scenario (compare FEP).
Severity	The severity of negative impact to one or more project values, also called consequences. Impact potentially arises from the occurrence of a scenario, or (in the case of a FEP) from the occurrence of the most important among a cluster of scenarios. Severity is semi-quantified on a categorical 1-to-5 scale. Where potential impact to more than one project value may occur, the severity quantity is defined with respect to the most-impacted value.

4.2 Initial Stage 1 Risk Assessment

Project Values

Project values are a fundamental risk assessment concept. Project values are of two types: 1) objectives to be reached; and 2) desirable present conditions to be maintained. Other terms for this concept include risk receptors, elements of concern, and risk targets. In order to evaluate degrees of potential impacts, it is fundamental that the receptors of potential impacts be specified. Project values are the specific entities that are considered to be at risk.

Table 3: Phase III SOPO Statements that Establish Project Values in 2008 (acronyms spelled out and minor edits for clarity)

Project Value	SOPO Statements
Health and Safety	The Phase III efforts will continue to refine the monitoring, verification, and accounting (MVA) techniques utilized in Phase II and incorporate new technologies to understand potential leakage pathways of the larger-scale test, provide post-injection monitoring, and provide assurance that health and safety requirements are fully taken into account.
Financial	By linking Phase II and proposed Phase III tasks, the project goal is to reduce costs and test large-scale sequestration sooner than might otherwise be possible.
	Infrastructure development will involve design of operating and injection systems, optimization studies to evaluate equipment capabilities and costs, and procurement of equipment.
Environment	National Environmental Policy Act activities will be included in [the Permitting Action] Plan.
	Illinois Environmental Protection Agency (EPA) guidelines for closing Underground Injection Control (UIC) sites will be followed. US EPA-recommended UIC Class V for CO_2 sequestration injection wells into brine water bearing formations. (Note that Class VI wells had not been designated until late 2010.)
	Initial site environmental monitoring is designed to obtain a baseline of environmental parameters.
Research	The MGSC Phase III large-scale injection test focuses on acceptance by the saline reservoir of the CO_2 (injectivity), [the] ability of the reservoir to store the CO_2 (storage capacity), the integrity of the seals and of the entire site, and the entire process of pre-injection characterization, injection process monitoring, and post-injection monitoring to understand the fate of the CO_2. These targets can be summarized under the general processes of modeling, capture, injection, and monitoring.
Industry Viability	The project will continue with a strong emphasis on capture and transportation during Phase III, as well as making continued and new efforts in the area of public outreach and education.
	One of the goals of this Phase III project is to understand all issues necessary to develop and operate a commercial-scale sequestration site completed in the Mt. Simon [Sandstone] in the Illinois Basin.

Risk Definition

Within industrial risk assessment, risk is typically defined as the product of the severity and likelihood (or probability) of an event that may impact one or more project values. This definition, already in use by Schlumberger, was adopted for the IBDP.

Severity and likelihood are typically semi-quantified through expert judgment on multi-step categorical scales, although in narrow contexts some components (such as financial impacts) can be quantitatively estimated. In some systems, additional attributes such as observability and treatability are factors in risk quantification.

Workshops included areas of technical consideration – geologic distribution and engineering, and areas of non-technical consideration – public engagement, legal and regulatory, infrastructure. In both cases project staff and outside experts participated in the assessment of associated risks with the category in which they were most familiar.

Risk Scales

Quantification using Relative Scales

In risk evaluation, quantification most often means that descriptions of risk and risk components are expressed in ranking or *ordinal* numbers (i.e., first, second, third, etc.) rather than measurable *cardinal* numbers *(1.725, 1897.2, etc.)*. This property is acceptable and useful for several reasons:

1) Ordinal numbers empower project managers to *order* (prioritize) concerns and actions;
2) Ordinal numbers allow simultaneously ranking priority across disparate scales, such as one scale involving environmental impact and another involving project schedule; and
3) Ordinal scales can be rapidly applied through expert judgment, without waiting for rigorous estimates of measured quantities.

Risk values can become rigorously quantitative if either: a) Large numbers of observations establish the recurrence frequency of a well-defined chain of events, or b) A well-defined scenario involving a well-defined system can be numerically modeled using well-validated software. Given that neither a) nor b) is the case for most risk matters that concern IBDP, all risk quantities addressed herein are expressed in ordinal numbers.

As noted above, risk is quantified here as the product of its components *severity* and *likelihood (*S^xL). Both severity (S) and likelihood (L) are quantified on five-category scales. As these scales involve a mix of judgmental and potentially observable or measurable criteria, they are overall categorical/ordinal. When risk evaluation was designed for IBDP, it was apparent that conceptions of typical or ordinary levels of severity and likelihood had to be distinguished from conceptions of worst case. Accordingly, quantities of upper-bound, best-guess, and lower-bound were defined for both S and L. Collecting six data points for each risk element proved to not only require great effort, but also add complexity to the data analysis. As detailed later in this report, collecting only three of these quantities proved to be a better practice, yielding similar quantifiable results. The three surviving risk component quantities are S_{ub} (severity upper-bound), S_{bg} (severity best-guess), and L_{bg} (likelihood best-guess), where L_{bg} is the likelihood that impacts having severity = S_{bg} will occur. No discrete likelihood value is (nor can be) associated with S_{ub}.

Severity

Against each project value, a range of potential negative impacts is expressed according to a categorical five-point severity scale (Appendix 7.1). The impact scale is categorical (as opposed to being a function of measured values) because many impacts (and the aggregate of multiple impacts) can be judged and expressed more consistently in relational words than in numbers. There are five categories because that scheme allows adequate impact differentiation while also allowing rapid and consistent judgment by subject-matter experts. Though other numbers of categories are used, five-point severity scales are common within industrial risk management. The purpose of the severity scale is to provide a consistent basis by which to compare impacts within a project value.

For the project values that are in typical industrial usage, the impact ranges and categorization are also typical. For the Research and Industry Viability values, category definitions were developed specifically for the IBDP. Depending on the specific scenario, an undesired event could have impacts to more than one project value: For example, the potential failure of CO_2 storage could entail environmental and/or health and safety impacts. For the IBDP, a financial impact of CO_2 lost from storage was also assumed, resulting from a hypothetical financial penalty under a carbon-price assumption of USD $30 per ton. Where impacts may affect multiple project values, the most severe potential impact defines the severity value.

For each project value, the category descriptions are only examples of relative impact magnitudes; they do not embrace the full variety of conceivable impacts. Categorizing potential impacts across multiple values is necessary for comparing risks and for rationalizing treatments. However, impacts of the same level in two different project values are viewed as equal only in the sense that they motivate similar levels of attention and effort in risk treatment.

Likelihood

The likelihood of negative impact is also categorized in five levels. Because actual experience with GS projects is not yet sufficient to estimate recurrence intervals for many pertinent events, experts were asked to estimate Likelihood based on their expectations relative to an arbitrary standard of *100 similar projects during 100 years* (Appendix 7.1). The planned project duration is ten years including three years of post-injection monitoring, but analyzing a longer time period is necessary so that potential longer-term effects are not ignored. The Likelihood scale presents a roughly categorized version of probability that provides a consistent basis for comparison.

In other projects and contexts, it is common to use arithmetic or roughly logarithmic scales of probability values, such as <1%, <3%, <10%, <30%, <90%, in place of the comparison (with a number of other hypothetical projects) that was used for the IBDP. Each type of scaling is arbitrary, and neither one is measurable, so the choice of scale type rests largely upon what system is most easily conceived and applied by the risk evaluators. During workshops and through spreadsheets, risk evaluators appeared to have applied the categorized Likelihood values with facility and consistency.

Risked Entities: FEPs and Scenarios

Various types of entities exist for which the probability and severity of negative impact (i.e., the risk) might be evaluated. At one end of the spectrum, the impact ranges and occurrence probabilities of narrowly defined events might be found in historical (actuarial) data gathered about occurrences identical to those that could occur at the project. Without actuarial data, experts can still make near-quantitative judgments about the probability and severity of relatively well-defined chains of events (scenarios). At the other end of the spectrum, risk levels can be holistically — yet more qualitatively — associated with broader concepts important to the project. For results to be applicable toward defining risk treatments, any project risk evaluation must define specific scenarios, but to provide the latitude for envisioning the critical aspect of unfamiliar risks, any project risk evaluation should also consider broad concepts. These may be considered as an initial step towards scenario evaluation or as check for thoroughness afterward. Risk identification for a new project should not be limited to evaluating a pre-defined list of scenarios.

Both scenarios and broad concepts were evaluated for the IBDP. To avoid missing sources of risk by focusing too narrowly, risk assessment at Stage 1 began with FEPs. FEPs evaluated by the large workshop group as having higher risk were used as the basis for writing scenarios. Risk levels for individual scenarios were monitored informally by individuals assigned as Responsible or Accountable for each scenario (see page 23). The Stage 2 evaluation was conducted using the cumulative project experience and a more specific list of scenarios for tighter focus risk evaluation. The following sections describe this progression in greater detail.

Stage 1 FEP List

The list of FEPs evaluated by each working group was selected from the FEP database created by Quintessa Ltd. for CO_2 sequestration projects (see www.quintessa.org), with some modifications and additions specific to the project. The Quintessa list captures most of the sources of risk that should be considered for a generic CCS project, but specific characteristics of any individual project are likely to extend the list. The IBDP FEPs list includes many FEPs typical of large-infrastructure projects such as vehicle hazards and chemical exposures, and also includes distinctive CCS-related FEPs in these areas:

- Reservoir porosity, permeability, and other geologic uncertainties;
- Induced seismicity;
- Groundwater contamination;
- Data access and archiving;
- Integration of surface- and subsurface-engineered components;
- Internal and external project communications;
- Regulatory developments; and
- Post-injection monitoring requirements.

Stage 2 Scenario List

The Stage 2 risk assessment exercise was designed to identify and evaluate the risk elements (FEPs or scenarios) considered important to further project progress, rather than to specifically re-evaluate all the risks from Stage 1. To begin the process, a set of scenarios (rather than FEPs) was selected. Operational experience at mid-project promoted a more detailed focus on project related risks reducing the potential to miss important general areas of risk. The Stage 2 scenario set was seeded from the Stage 1 scenarios that remained active as risk sources (37 of 94 scenarios), then supplemented by 96 additional scenarios gathered during a pre-workshop process focusing on the changes in the project environment and knowledge (Table 4). This combined list of 133 was refined to a total of 83, with two scenarios added during the workshop, for a total of 85 scenarios discussed and evaluated for Stage 2.

The risk evaluation process was designed to be responsive to project needs in terms of both workflow efficiency and applicability. Due to the formative nature of the risk assessment process, no direct comparison of risk levels from Stage 1 to Stage 2 is possible. In future commercial projects repeat risk evaluation is recommended. Key factors to consider are potential changes to scenarios over time, identifying areas of persistently higher risk, and investigating the effectiveness of various risk treatments.

Initial Risk Evaluation Workshops

Timing and Design

Initial risk evaluation was conducted through two face-to-face workshops (Stage 1, Parts 1 and 2) held at the offices of the Illinois State Geological Survey (ISGS) in Champaign, Illinois. Each of the two Stage 1 workshops incorporated mainly internal experts, plus a small number of external experts and representatives of the public.

The first of the two Stage 1 workshops was conducted February 19-21, 2008, in conjunction with the general project kickoff meeting. The kickoff meeting was attended by project participants from all member organizations, and a broad range of expertise was available for both evaluation workshops in Stage 1. The second of the two Stage 1 workshops was held July 15-16, 2008. Most of the participants involved in the two Stage 1 risk workshops visited the project site and contributed to a half-day's sharing of information on project data, plans, and risk-assessment methods.

During the project's early months, about 25 individuals met for two-to-three days to carry out the initial risk workshops, and approximately 15 additional person-days were spent by a smaller risk assessment team. The Schlumberger Carbon Services Risk Manager worked roughly one-third to one-half time on the IBDP during a three-to-four month period to develop the assessment. Site information and plans were presented by the participants most familiar and involved with the project and its physical and cultural setting. Presenters compared well-known site features with those less well-known, and identified uncertainties with potentially important consequences.

Training in risk-assessment techniques and procedures included a discussion of heuristics (rules of thumb that individuals apply, often unconsciously, when they make decisions under uncertainty). Although paying attention to heuristics and biases, judgment under uncertainty, and risk perception is important to workshop design, the pursuit of precise and repeatable expert-elicitation data through rigorous techniques is constrained by practical resource limits and human patience. The Stage 2 Workshop later in the project made improvements in this regard (see Changes in Technique, page 19).

Participants and Working Groups

Risk evaluation input was solicited from a wide range of experts mostly from within the project's member organizations. These experts were organized into six working groups defined by professional discipline. Each group was asked to evaluate a set of FEPs within its areas of expertise. The first of the Stage 1 workshops consisted of three working groups and met immediately after the project kickoff meeting in February 2008. Stage 1 Part 1 focused on subsurface, near-surface, and airspace areas near the planned injection well site. A specific objective was to define the MVA Program using risk information.

The second of the Stage 1 workshops consisted of three working groups that met during June 2008. The Stage 1 Part 2 workshop focused on the surface infrastructure (CO_2 compression-dehydration-delivery system), the neighboring community, and an array of ownership and environmental topics including finance and permitting. The groups' spheres of consideration were to some extent overlapping, and many topics and concepts were considered by more than one working group. Several individuals participated in both Stage 1 workshops.

4.3 Pre-Injection Risk Evaluation Stage Gate

The evaluated, risk-ranked FEPs provided the basis for developing specific scenarios, which subsequently provided the basis for developing risk treatments. Early in the project, about 85 of the 201 risk treatments were flagged as *required for completion before injection start*. As the IBDP made progress toward the start of injection, many of these actions were documented as complete by their respective *Responsible* parties, and verified as complete by their respective *Accountable* parties. For further detail on the Reasonable, Accountable, Consulted, and Informed (RACI) process, see page 23.

When the project received its injection permit, about 25 of the defined risk treatments remained either known to be incomplete, or believed to be complete, but undocumented as such. Accordingly, the status of these risk treatments and associated risk-bearing scenarios was reviewed by the Principal Investigator, supported by the Schlumberger Carbon Services Project Manager, and Risk Manager. It was determined that the risks addressed by the outstanding response actions did not pose immediate threats to safety and/or had been adequately addressed through other actions. Consequently, residual risk was evaluated as being acceptable, and CO_2 compressor start was authorized. More detailed description of risk treatment and treatment tracking is provided in Risk Treatment and Risk Tracking (page 23).

4.4 Mid-Project Stage 2 Interim Risk Assessment

By the spring of 2013, more than a year of operational injection experience had been gained, and substantial additional project data were available. The new data consisted of several rounds of sampling results from the in-reservoir observation well, continuous pressure and temperature data at multiple surface points and depth levels, processed 3D vertical seismic profile (VSP) data from which the distribution and movement of the subsurface CO_2 plume was interpreted, and other data sets and operations records. All of this information expanded the basis for evaluating risk at a greater level of knowledge than existed at project start. Accordingly, risk was thoroughly re-assessed during February 2013. In contrast to the summary risk evaluation that was performed as a stage gate in advance of the November 2011 decision to start CO_2 injection, the February 2013 Interim Risk Assessment was intended to revisit first principles. The 2013 workshop acknowledged the five project values, reviewed the project status changes since 2008, discussed established and emergent sources of risk, developed treatments to reduce risks, and created the information basis for a reformulated risk management plan.

In general terms, the objectives of the Stage 2 mid-project interim risk assessment were the same as those in the early-project Stage 1 evaluation: to identify and evaluate sources of risk, to plan actions to reduce risks that are deemed unacceptably high, to establish practices that will maintain risk at acceptable levels, and to document risk treatments and risk status.

In the Stage 2 evaluation, key differences from the focus of the Stage 1 evaluation included:

- The project was no longer in the initiation and design phase, but was within a phase of continuous and relatively steady operation.
- The elements whose risks were evaluated were well-defined scenarios, rather than conceptual FEPs. Operating experience provided confidence that project risks could be adequately conceived and captured using a manageable number of scenarios -– while acknowledging that the list of important scenarios must always be open to new entries.
- Risk treatments trended away from tailored and finite solutions to evident problems or process gaps, and trended toward ways to maintain vigilance and to progressively improve project operations.
- The central functional object of risk management, and of the project as a whole, was no longer the successful initiation of CCS system operations. Rather, it has become the maintenance and documentation of those operations and their effects, so that the conditions for safe project closure and cessation of monitoring can later be proven.

4.5 Comparison of Stage 1 and Stage 2 Risk Assessment

Changes in Purpose and Scope

The Stage 2 risk interim assessment exercise was designed to identify and evaluate the risk elements considered important to further project progress, rather than to specifically re-evaluate the elements from Stage 1.

To begin the process, a set of scenarios (rather than FEPs) was selected; advanced knowledge at mid-project promoted a more acute focus with little potential that an important general area of risk would be overlooked. The scenario set for Stage 2 was seeded from the Stage 1 scenarios that remained active as risk sources (37 of 94 scenarios), then supplemented by 96 additional scenarios gathered during a pre-workshop process focusing on the changes in the project environment and knowledge; potential sources of change and example scenarios may be found in Table 4. This combined list of 133 was refined, and two scenarios added during the workshop, for a total of 85 scenarios discussed and evaluated for Stage 2 (Appendix F). The Project Values, definition of Risk quantity (S^xL — Severity times Likelihood), and Severity and Likelihood scales remained the same as for Stage 1. Scenario topical groupings differed to some extent because of the advance in project phase.

Table 4: Sources of change at Stage 2.

Change Area		Description of Potential Changes	Example Scenario
1	Goals Changes in project goals or objectives would affect risks and their relative importance.	New objectives may be established by the project or the US Department of Energy (DOE). Goals may be more quantifiable, specific, and/or stringent.	New emphasis on safe CO_2 surface operations imposes limitations that impede subsurface monitoring.
2	Project Phase Design, construction, and startup are complete. Focus is on steady operations and meaningful observations.	Original guidelines and procedures may no longer be suitable. Original focus on safety may have slipped. Monitoring may not be building adequate data toward closure. Financial resources may be inadequate to accomplish the mission.	Field staff becomes less vigilant; an unsupervised project subcontractor vehicle collides with field equipment.
3	Knowledge There is greater operating experience and site knowledge, including seeing actual effects of CO_2 injection.	Improved knowledge from physical samples, reflection seismic, microseismic, etc., may improve the understanding of relationships between observations and risk levels.	Responsibilities pass to new personnel; data or learnings are lost while experience is regained.
4	Physical Setting There is now pressurized CO_2 in the ground, brine has moved, and rock stresses have changed.	Non-project CO_2 will be injected nearby. Significant land-use changes may occur, which could affect access and/or the ability to monitor, or might create signals that interfere with monitoring and/or interpretation.	Simulations show that injection by a second project is likely to cause the IBDP plume to cross a property boundary after IBDP injection is complete.
5	Socio-Economic Setting Informal rules or operating constraints or neighbor relationships might have changed.	Social influences, funding environment, climate policy, politics, and international norms may change. Effects of stakeholder interactions may be positive or negative. The social environment may become collaborative or conflict-oriented.	Fear of fracking incites strong reaction to microseismic activity; project is ordered to stop injection until it proves safety.
6	Regulatory Environment Formal regulations or permit requirements may differ from the initial assumptions or conditions.	Operating rules or closure criteria may change. Relationships with regulatory agencies may change.	Project is ordered to stop injection until demonstrating compliance with Class VI rules.
7	Opportunity Given all of the changes over time, a new goal may have become feasible and/or desirable, or an old one infeasible.	The adjacent project could expand the monitoring opportunities or the range of effects that could be tested. The Sequestration Training and Education Program (run by the MGSC) or the programs at the Richland Community College may open opportunities to advance the viability of the CCS industry. The *usage* portion of CCUS may open new research objectives for the IBDP.	A key monitoring component stops working; not enough injection time remains to establish trends through another method and closure is delayed.

Changes in Technique

The effectiveness of discussion and the efficiency of data acquisition for the risk assessment were improved in Stage 2 over the techniques applied in Stage 1. These improvements were enabled by using wireless keypads and related software for real-time data capture during the risk workshop. At Stage 1, participants had been divided into small groups according to area of professional expertise, and FEPs were assigned topically and separately (with some overlap) to each small group. Accordingly, each individual participant deliberated and contributed with respect to only a subset of the total number of risk elements. At Stage 2, all scenarios were discussed and evaluated in plenary session, yielding a broader exploration of potential risk pathways and a much greater number of quantitative evaluations. In contrast to Stage 1, where the adequacy of each individual's expertise for evaluation was assumed based on sub-group assignment, personal expertise at Stage 2 was tracked using scaled self-assessment within each of the scenario topic groups. Self-rated expertise (per scenario) was applied during data evaluation, distinguishing expert and non-expert views.

Comparison of Risks between Stage 2 and Stage 1

General

A sense of the progression of relevant risk issues can be gained by comparing, for similar topic groups, the concentrations of scenarios and treatments between the two stages. Risk levels cannot be directly compared on a calibrated basis because the elements evaluated in a workshop setting at Stage 1 were FEPs while those at Stage 2 were scenarios. At Stage 1, 22% of the >200 risk treatments involved well construction, well completion, or reservoir geology; by Stage 2, the risks associated with these treatments had largely been resolved and were represented by no scenarios or treatments. In contrast, based on the distribution of about 100 Stage 2 scenarios, risks arising (or more highly emphasized) at that later point included those associated with potential felt seismicity, plume footprint, permitting (because of uncertainty about Class VI status), knowledge capture and propagation, and mechanical integrity of instrumentation. Risk topic areas receiving similar emphasis at both stages included budget, health and safety, and public interactions.

Top-Ranked Risks

Stage 1

At Stage 1, the fundamental entities evaluated for risk were FEPs. Because the workshop was structured in breakout groups, and due to uneven post-workshop spreadsheet response, individual FEPs received a varied number of numerical evaluations representing varied levels of expertise. Therefore, FEP ranking was determined based on plurality of high-risk values (number of scores of 20, number of 16s, etc.) rather than on a straight average of values. This inconsistency of the captured data added uncertainty to the analysis, and was among the reasons that methods were modified for Stage 2.

The 19 top risks shown in Table 5 include all FEPs that received at least two risk scores of 12 or greater. The complete ranked Stage 1 FEPs list is provided in Appendix 7.2.

Table 5: Top-Ranked FEPs, Stage 1

RANK by plurality of high-risk scores	FEP	MAX RISK	AVG RISK
1	Toxic geologic components (metals)	20	7.2
2	Contamination of groundwater by CO_2	20	5.8
3	Undetected features	16	7.1
4	Human activities in the surface environment: on site	16	7.5
5	Exogenous economics: Supply prices	15	9.8
6	Near-surface aquifers and surface water bodies	15	5.6
7	Accidents and unplanned events: Project	15	5.8
8	Community characteristics	15	5.0
9	Legal/regulatory: Property rights and trespass	12	6.9
10	Fractures and faults	12	7.8
11	Schedule and planning	12	6.7
12	Reservoir pore architecture	12	6.2
13	Reservoir geometry	12	5.6
14	Actions and reactions: Local community	12	6.6
15	CO_2 solubility and aqueous speciation	12	8.0
16	Seismicity (induced earthquakes)	12	6.1
17	Actions and reactions: National/international Special Interest Groups and Non-Governmental Organizations (NGOs)	12	5.5
18	Legal/regulatory: Construction, discharge, and other operations permits	12	5.0
19	Land and water use	12	4.4

The Stage 1 top-ranked risks focused on:
a) Geologic uncertainty;
b) Operational uncertainty;
c) Legal/Regulatory uncertainly; and
d) Social uncertainty.

During the first year of CO_2 injection, concerted efforts to manage risks and to reduce uncertainties was undertaken and substantially decreased concerns in a) and b), and d) above.

Approximately the top half of the Stage 1 risk-ranked FEPs list was used to generate scenarios. For the purpose of risk ranking, each scenario was initially assigned the value of its highest-risk associated FEP. Appendix 7.2 shows each Stage 1 scenario, its scenario group, and the risk rank of its top associated FEP. After treatments were developed for each scenario, the scenarios were topically grouped, which in practice meant grouping them according to their principal risk treatments. The 29 treatment groups are shown in Table 6, along with the number of treatments assigned to each group.

Table 6: Stage 1 Risk Treatment Groups. (also used to group scenarios)

Group ID	Stage 1 Treatment Group	Count of Treatments
OE01	QHSE plan: Subsurface and seismic operations	6
OE02	QHSE plan: Surface and injection operations	12
OE03	Site security plan	1
OE04	Drilling plans, deep wells	7
OE05	Completion plan: in-zone wells	4
OE06	Completion plan: Geophone wells	5
OE07	Completion plan: Shallow groundwater monitoring wells	5
OE08	CO_2-resistant well construction	4
OE09	Injection operations envelope	16
OE10	Injection operations and shut-in plan	10
OE11	Equipment sparing plan	4
OE12	Operational monitoring plan	9
PM01	Overall risk management	10
PM02	Permits	8
PM03	Data management plan	8
PM04	Budget	12
PM05	Staff and coordination	13
PM08	Communications policy	7
PM12	Injectivity backup plan	1
PM13	CO_2 movement contingency plan	9
PM14	Goal triage	2
PM15	Communications plan	13
SR01	Site selection and area of review	9
SR02	USDW baseline chemistry	3
SR03	Reservoir characterization plan: in-zone wells	8
SR06	MVA plan	12
SR08	Surface seismic	3
SR09	Static earth model	2
SR10	Simulation plan	3
		206

Stage 2

At Stage 2, the plenary workshop discussed and evaluated each scenario for risk. This structure produced a larger and more consistent data set from which to calculate average risk. All scenarios are shown in Appendix 7.6. The 14 scenarios shown in Table 6 include the top ten scenarios overall by average risk value, and an additional four scenarios that were ranked highest within their scenario group.

Prominent among the Stage 2 high-ranked risks are induced-seismicity scenarios. Shortly before the Stage 2 workshop, field data had begun to demonstrate patterns in microseismicity, and intensive work to understand connections between injection operations and microseismicity was under way.

Other themes prominent among the top risk-ranked Stage 2 scenarios include concerns about the footprint of the CO_2 plume, continued proper function of stressed mechanical and instrumental components through project end, and regulatory or public interactions that could present institutional roadblocks to project completion. When compared with Stage 1 scenarios and FEPs, Stage 2 evaluation showed that while concerns about potential physical impacts from injection had generally receded along with the acquisition of project data and operating experience, risks related to public concerns are still judged important. In addition, Stage 2 scenarios and evaluations demonstrate forward-looking focus on eventual project closure.

Table 7: Top Risk-Ranked Scenarios, Stage 2

Scenario Group	Scenario	Avg Risk
Seismicity	CCS2 increases reservoir pressure, triggers a felt seismic event, and regulators shut down both projects pending investigation.	9.71
Plume Footprint	Plume migrates beneath sensitive area or unexpectedly far, increasing monitoring requirements and cost.	9.17
Regulations, Permitting, Closure	Regulatory agency is surprised to learn of a connection between seismicity and injection, and requires shutdown pending investigation and additional monitoring.	8.65
Regulations, Permitting, Closure	CCS2 logs show a fault cutting the Mt. Simon that looks important (as a potential source of seismicity or influence on fluid movement), and regulators require IBDP injection to stop.	7.94
Plume Footprint	New and untested technologies malfunction, increasing cost, and impairing data acquisition.	7.94
Data Interpretation and Care	The IBDP project data is required to be made public without time for adequate analysis and/or significant publications from the project team, resulting in misrepresentation of the information.	7.75
Seismicity	ICCS does not effectively apply IBDP research on microseismicity, and induces seismicity that causes regulators to shut down both projects.	7.65
Operations, Mechanical Integrity	Packer in CCS1 fails and a costly workover is needed.	7.53
Staff and Expertise	A valuable subcontractor has scarce resources and does not send appropriate staff levels to complete a job.	7.53
Seismicity	IBDP operations cause seismic event that is felt by people in Decatur, leading to news reports that CCS causes earthquakes.	7.50
Health, Safety, Environment	Injury from a common industrial or drilling hazard.	7.29
Environmental Monitoring	Westbay multilevel groundwater characterization and monitoring system fails beyond repair, and VW1 must be re-completed.	6.89
Public interactions	An unplanned event occurs; news media become involved; key people are unavailable but a public response is needed.	6.50
Budget, Cost	DOE funding is reduced, and not enough funds remain for proper site/project closure.	5.94

4.6 Risk Treatment and Risk Tracking

General

Risk management flows from risk identification to assessment to evaluation to treatment.



Figure 1 – Risk Management process

A risk treatment (internally called a *risk response action* during Stage 1) is intended to lower risk associated with a well-defined scenario by reducing either the likelihood that the scenario will occur, the severity of consequences if it does occur, or both. As an example, for a scenario involving injury caused by mechanical failure of energized or rotating equipment, one risk treatment would be to improve the equipment quality to reduce the likelihood of failure; another would be to ensure that personnel wear adequate protective gear. Contingency plans and other preparations for unexpected or uncommon events constitute risk treatment through severity reduction.

In theory, the effectiveness and resource efficiency of each treatment in reducing its target risk(s) can be catalogued and up-to-date risk status and treatment progress can be aggregated from bottom-up data whenever desired. In practice, this high degree of monitoring demands considerable internal resources. Consequently the approach now taken is periodic summary reporting at a relatively generalized level (see Risk Reporting following), augmented by more rigorous and quantitative stage-gate-oriented review.

RACI Definition of Responsibilities in Risk Management

A common way for project managers to assign responsibility for project activities is within a RACI matrix. In this respect, risk treatment activities are no different, and the RACI tool has been applied. This standard project management acronym indicates persons or entities (individuals or groups) that are Responsible, Accountable, to be Consulted with, or to be Informed about an action. In Stage 2, IBDP uses the modified acronym ROCI, where O indicates Oversight. The Responsible assignment is for those who do the work to accomplish the task. Accountable (Oversight) is assigned to those who are answerable for ensuring that the task is completed correctly and thoroughly; the individual or group accountable also delegates the task to those who will be responsible to do the work. Consulted parties are those whose views and input are wanted, typically subject matter experts. Informed are those persons or entities that are kept up-to-date on the progress.

Appendix 7.5 shows the Responsible (R) and Accountable (A) assignments by role, for Stage 1. In Stage 1, RACI assignments were made for groups of risk treatments. In Stage 2, RACI assignments are instead grouped by scenario. The changed structure for Stage 2 signals that the fundamental responsibility of the Responsible person is to monitor the risk that is related to each scenario, which in turn requires tracking (or personally executing) the treatments associated with each scenario. Although complexity is unavoidable in either RACI structure, the Responsible assignment by scenario group more clearly supports a culture of risk awareness.

Note that in the example scenario in the previous section (see General above), the two proposed treatments might be executed by different persons, but the R person is expected to judge the effectiveness of both treatments toward reducing scenario risk. Aided by the project Risk Manager, the R person provides (via input to a spreadsheet or database) the information that tracks treatment progress and scenario risk status, and this information is subject to review by the A person. The R person is also charged to maintain awareness for emergent (new) risks within the topic area.

The key assignment of the Accountable or Oversight person is to review the effects of risk treatment (the status of risk) at significant project stage gates. At these stage gates or decision points, the Accountable persons respond to the Principal Investigator's need to confirm adequate risk control before proceeding.

During Stage 1, multiple formats of tracking spreadsheets were attempted, none of which was entirely satisfactory. The spreadsheets were kept as shared files on a server that was accessible internally within the project, and nominally available at all times. Generally, the R and A persons found it relatively time-consuming and burdensome to provide the requested information on both scenarios and treatments, resulting in less-than-desired data input. Consequently the information base did not serve project managers with as much currency as desired.

Early efforts were made toward developing a risk management database to efficiently process the inherent many-to-many logical relationships; regardless of its potential merits, the overhead cost of developing and fine-tuning such a database was prohibitive and ultimately beyond the scope of the project budget.

During Stage 2, the tracking spreadsheets have been simplified so that information on all scenarios is maintained on one page, and all information on treatments is on another page. The Risk Manager retains the updated file, and requests periodic updates from the R persons through verbal reports or submitted on spreadsheets. This more hands-on and less spreadsheet-centered process has achieved better balance between effort and information currency.

Risk Reporting

At the inception of IBDP, the two major industrial partners (Archer Daniels Midland Company (ADM) and Schlumberger Carbon Services) were able to draw upon long-established corporate standards for risk management, including regular risk and incident tracking and reporting. Similarly, the ISGS was following risk management procedures established by the University of Illinois. In 2008, Schlumberger lead these entities (as well as the company contracted to drill the planned injection well) in creating a bridging document whose primary purpose was to define HSE-policy primacy and to coordinate safety procedures across the partner organizations. Schlumberger procedures, for example, have been followed in respect to well drilling activities, while ADM procedures control site access and establish safety screening and training requirements for all personnel admitted to ADM property (i.e., the entire site). Both ADM and Schlumberger maintain ongoing safety incident reporting according to their established procedures: For ADM, reporting is required of any incident involving ADM property or staff; for Schlumberger, reporting scope includes anything involving the project itself, property or staff of Schlumberger, or subcontract project staff.

Schlumberger reports incidents involving both actual and potential losses internally so that the information gathered can support preventive risk reductions; information on Occupational Safety and Health Administration (OSHA) reportable incidents is also provided to ADM. All pertinent data is maintained in electronic databases.

At the start of Stage 2, a one-page internal project risk report was initiated, and was carried through the completion of the planned CO_2 injection. The Risk Manager prepared this report about every two months, highlighting current risk issues and efforts toward risk reduction. The report was based on current updates from the R persons, as captured by the risk manager in the tracking file previously described. Like the scenario-based structure of risk treatment tracking, this instrument contributed to maintaining the culture of risk awareness throughout project life.

At mid-project, the updating of risk information for the IBDP was carried out on an approximately bi-monthly basis. Each Responsible person provided an assessment of risk status of identified scenarios, progress on risk treatments, and emergence of new risks (if any) within his or her respective topic area, and the risk manager maintained the accumulated risk information. Reporting to the IBDP occurred until injection ceased by way of bi-monthly one-page reports, and was supplemented as needed as risks changed. It is anticipated that as the project nears the end of injection and the subsequent cessation of monitoring, developing regulatory stipulations will help to shape the ways in which end-of-project risks are identified and evaluated.

4.6 Toward Closure: Future Stage Gates and Assessments

As highlighted during scenario capture and screening for Stage 2, project focus — and therefore the focus of risk management — is now centered on steady operation and progress toward closure. Stage gates and milestones where risk is to be re-evaluated should be defined with this in mind. Such time points might include:

- Following the final planned 3D seismic survey. This time point would provide the most complete data set to judge risks related to remaining technical uncertainties about the subsurface plume position and evolution. Although other data will continue to be collected, the latest groundwater sampling data and a relatively complete set of all other observations such as pressures, microseismicity, and data from shallow wells will be critical for interpretation.

  Risk evaluation based on the most complete attainable data set will initiate or fully constitute the fundamental end-of-project evaluation. Results of this evaluation will be processed and relied upon to establish with reasonable certainty whether the plume is sufficiently stable that monitoring may be safely discontinued.

- Before the decision to permanently cease injection. This time point is especially important for risk re-evaluation if contemporaneous abandonment of wells or of monitoring instrumentation is planned. Brainstorming-level risk identification should envision whether these actions could cause or increase some risks, and whether there might be reason (such as improved signal for a monitoring method) to continue injection for some longer period.

- In relation to any important change in project environment. Risk re-evaluation at this time point would be indicated in the event of startup of an adjacent non-IBDP injection (Illinois Industrial Carbon Capture and Sequestration Project - ICCS), a change in permit requirements, a locally felt earthquake, or a change in project funding.

5. Risk Management Lessons Learned

5.1 Risk Assessment Processes

The main process changes and Lessons Learned in the progression from Stage 1 to Stage 2 were:

1. Discussion and evaluation were held with all evaluation participants present (in plenary session), rather than in breakout groups by discipline. As risks can easily be missed in the gaps and overlaps between project components, between project functions, among partner organizations, and among professional disciplines, this format proved to be superior.

    - Lesson Learned: Discussion and evaluation in whole-group (plenary) session was preferable to breakout sessions, as it led to fully involving all experts, airing a wider range of views, and promoting greater discussion among professional disciplines.

2. The qualifying expertise for evaluating risks within topic areas was determined not through breakout-group assignment, but by self-rated expertise level. During data analysis, each severity or likelihood value chosen was correlated with the expertise level of the person who provided it. This enabled the identification of instances in which expert views may diverge from those of well-informed non-experts.

    - Lesson Learned: Self-rating of expertise level led to a more efficient means to understand how expert views diverged from those of well-informed non-experts.

3. Using a wireless keypad system made changes #1 and #2 (above) practical. The system recorded all severity and likelihood values and instantly displayed results on screen, flagged potential needs for further discussion, was more engaging for participants, much faster, and was error-free in archiving data.

    - Lesson Learned: A wireless keypad system for instantaneous feedback and display was beneficial and had the added advantage of archiving data easily.

4. Three risk quantities, not six, were collected per risk element. Data were collected strictly in the order $S_{ub} > S_{bg} > L_{bg}$ (see Risk Scales, page 29, for explanations of these quantities). The three-value quantification was clearer for participants and simpler to analyze, and it more clearly distinguished the upper-bound or worst-case severities.

    - Lessons Learned:
        1) Three risk quantities per element are adequate and easier for participants to analyze;
        2) Scenarios with very high worst-case severities must be treated differently from scenarios whose high risk results from higher likelihood.

5. Spreadsheet follow-up was used to complete the workshop evaluation; it became clear that the desire to distinguish group vs. individual evaluations was an unnecessary complication.

    - Lesson Learned: There is no need to distinguish group vs. individual in evaluations.

5.2 Risk Information Management

1. In addition to process changes for risk assessment, project experience has shown the need for improved information management and internal reporting. Although not yet fully achieved, the key goal for risk information management is a simple, easy-to-manage strategy.

    - Lessons Learned:

        1) Maintaining staff focus on actions helps to reduce and control major risks;
        2) Monitoring lesser and changing risks provides the basis for elevating them if necessary; and
        3) Incrementally accumulating the data to support risk-based project decision-making at key turning points is beneficial to project success.

2. Risk information is dominated by many-to-many relationships: Every scenario inevitably involves multiple FEPs; one FEP can generate multiple scenarios; one scenario may have several risk treatments; and one treatment may reduce risk from multiple scenarios. Topical groupings are also non-unique: A scenario that seems clearly to fall within one topic group may have its most important treatment within a different topic group, as would be the case for a scenario involving a monitoring activity that could incite public opposition to the project.

    - Lesson Learned: While flat spreadsheets are ideal for handling one-to-one and one-to-many relationships, the many-to-many relationships typical of risk information are often better handled using a relational database. It should be noted that inputs and outputs from such databases are still two-dimensional spreadsheets.

3. Changes made in risk information management since project commencement have reduced the amount of administrative time required. Because current tools have not satisfactorily supported the real-time collection and application of risk information, developing a process for ongoing, yet minimally burdensome participation, is the key to engaging project staff.

    - Lesson Learned: It is recommended that future multi-partner CCS projects consider a) Adopting a risk information management system already in use by one partner (if a suitable system exists); or b) Licensing such a system from a commercial vendor.

5.3 Ongoing Risk Management

Along with the area of risk information management, ongoing risk management shares the challenge of producing usable and critical results while adding as little as possible to project overhead. While command-and-control approaches are suitable for some easily defined high-exposure activities, they can add greatly to staff burdens if applied to all risks, and can increase some risks if the approach demands constant attention. The appropriate level and style of attention is not easily understood nor reached.

Compared to a concrete project management topic like well construction, or to a topic demanding constant attention like budget, risk can be challenging to grasp and to focus on. Efforts to collect, evaluate, manage, and deliver risk information build the framework for successful risk management, but in the first instance, these elements must serve to motivate project staff to constantly envision project risks and risk remedies.

While the spreadsheets, checklists, and other tools of the trade are useful, the most important product of a successful risk management program is establishing and maintaining a *culture* of risk awareness and risk reduction.

6. Conclusion

A Risk Management process was applied to the IBDP that helped to identify and treat risks. The process was iterative and adaptive as more was learned about CCS. Three risk workshops were conducted in the IBDP, two of which were prior to injection and one during the injection phase. The risk workshops and risk management techniques helped the project to operate safely and achieve its objectives.

To date, the IBDP safely stored one million tonnes CO_2 and is, at the time of this report, in the post-injection monitoring phase. The lessons learned from IBDP have been and continue to be applied to subsequent CCS projects. Much of the success of the IBDP can be attributed to the culture of safety and risk awareness that was cultivated and facilitated in all aspects of the project by the personnel involved.

7. Appendices

7.1 Risk Matrix, Project Values, and Risk Scales

Hazard Analysis and Risk Control;

-25 to -20	BLACK	NON-OPERABLE:	Evacuate the zone and or area/country
-16 to -10	RED	INTOLERABLE:	Do not take this risk
-9 to -5	YELLOW	UNDESIRABLE:	Demonstrate ALARP before proceeding
-4 to -2	GREEN	ACCEPTABLE:	Proceed carefully, with continuous improvement
-1	BLUE	NEGLIGIBLE:	Safe to proceed

MITIGATION

Control Measures

PREVENTION

LIKELIHOOD

SEVERITY

		Improbable 1	Unlikely 2	Possible 3	Likely 4	Probable 5
Light	-1	-1 1L	-2 2L	-3 3L	-4 4L	-5 5L
Serious	-2	-2 1S	-4 1S	-6 3S	-8 4S	-10 5S
Major	-3	-3 1M	-6 2M	-9 3M	-12 4M	-15 5M
Catastrophic	-4	-4 1C	-8 2C	-12 3C	-16 4C	-20 5C
Multi-Catastrophic	-5	-5 1MC	-10 2MC	-15 3MC	-20 4MC	-25 5MC

White arrow indicates decreasing risk

PROJECT VALUE	Statements and topics in Phase III Objectives that establish this value	Explanation
Health & Safety	*The Phase III efforts will ... provide assurance that health and safety requirements are fully taken into account.*	self-evident
Financial (USD)	*... our goal is to reduce costs and test large-scale sequestration sooner than might otherwise be possible. ...* *Techno-economic analyses will be applied using the emission and injection data and expense data for the large-scale CO2 sequestration process at the ADM site.*	(1) Budget exceedances threaten achievement of project goals that occur later in time. (2) Besides project budget impact, higher costs impair the ability of CCS to ameliorate ghg's.
Environ -ment	*Well UIC permitting, NEPA compliance, compliance with other applicable environmental regulations.*	self-evident. Baseline environmental monitoring to include ground water and soil gas sampling.
Research	*The MGSC Phase III large-scale injection test focuses on **acceptance** by the saline reservoir of the CO2 (injectivity), ability of the reservoir to **store the CO2** (storage capacity), the **integrity of the seals** and of the entire site, and the **entire process** of pre-injection characterization, injection process monitoring, and post-injection monitoring to understand the fate of the CO2.*	Four distinct and articulated project goals are expressed by the need to understand Injectivity, Capacity, Storage Integrity (Effectiveness), and the Process of data acquisition, interpretation, and longterm storage.
Industry Viability	*We will continue with a strong emphasis on capture and transportation during Phase III, as well as make continued and new efforts in the area of public outreach and education. ... to ensure sufficient outreach is conducted in the field site area and to build informed and supportive constituencies. ...* *One of the goals of this Phase III project is to understand all issues necessary to develop and operate a commercial scale sequestration site completed in the Mt. Simon in the Illinois Basin.*	The ability of CCS to reduce ghg's depends on maintaining public confidence in (1) The feasibility and safety of the technology, and (2) The organizations (private and public) who execute and monitor this work.

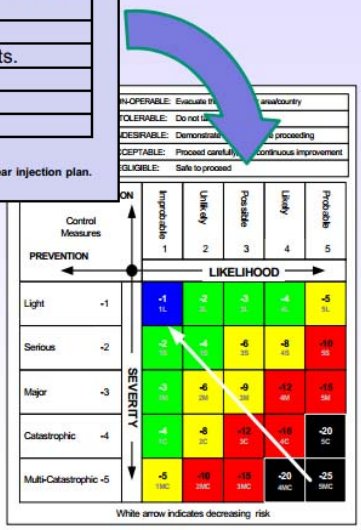
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Likelihood of Impact

... during ILADM Project Storage Time Horizon (assume 100 years)

		If there were 100 projects like ILADM*, impact related to this risk element (FEP) would occur ...
Improbable	1	... probably not at all; never.
Unlikely	2	... fewer than three times among the 100 projects.
Possible	3	... 5 or 10 times among the 100 projects.
Likely	4	... in around half of the 100 projects.
Probable	5	... in most or nearly all of the projects.

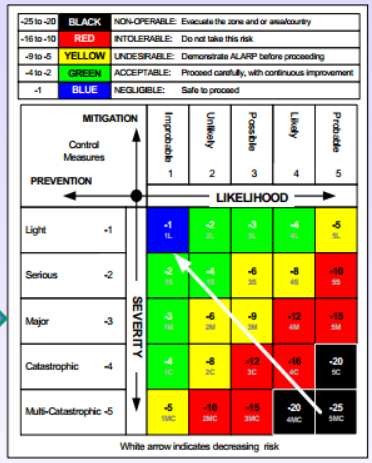
*** Similar setting, similar levels of knowledge and uncertainty at this stage of the project, same 3-year injection plan.**



2

Severity of Impact		PROJECT VALUES that may be at risk (bold: SLB Severity Matrix standard. *Italics*: proposed standard.) Health & Safety	Financial (USD); *escaped tons CO₂ @ $30/t*	Environment	Research	Industry Viability
Light	-1	Minor Injury or Illness, First Aid	<10K$ *<333t*	Discharge < reporting thresholds; Hazmat Spill <100 Liters; Produced Water Spill <50 Barrels	*Little or no progress toward 1 of 4 goals.*	*Project Lost Time >1day. Moving-vehicle citations.*
Serious	-2	Temp. Disability, Hospital to 1 day, Lost Days 1-100	10-100K$ *333-3333t*	Discharge > reporting thresholds; Hazmat Spill <1000 Liters; Produced Water Spill <250 Barrels	*Little or no progress toward 2 of 4 goals.*	*Project Lost Time >1 wk. Regulatory Notice without fine. Local allegations of unethical practice or mismanagement.*
Major	-3	Perm. Disability, Lost Days >100, Intensive Care >1 day	100-1000K$ *3333-33,333t*	Discharge causes area evacuation or wildlife loss; Hazmat Spill <10K Liters; Produced Water Spill <500 Barrels	*Little or no progress toward 3 of 4 goals.*	*Project Lost Time >1 mo. Permit suspension. Majority local opposition or substantial negative local media coverage.*
Catastrophic	-4	Fatality	>$1,000,000 *>33,333t*	Uncontrolled release of radioactive matl.; Hazmat Spill >10K Liters; Produced Water Spill >500 Barrels	*Little or no progress toward 4 of 4 goals.*	*Project Lost Time >1 yr. Int'l media coverage of law violations, questionable ethical practices, or mismanagement.*
Multi-Catastrophic	-5	Multi-fatality	*>333,333t*	Multi-Catastrophic	*No gain in understanding applicable to future projects*	*Negative public experience results in legal ban on similar projects.*

-25 to -20	BLACK	NON-OPERABLE: Evacuate the zone and or area/country
-16 to -10	RED	INTOLERABLE: Do not take this risk
-9 to -5	YELLOW	UNDESIRABLE: Demonstrate ALARP before proceeding
-4 to -2	GREEN	ACCEPTABLE: Proceed carefully, with continuous improvement
-1	BLUE	NEGLIGIBLE: Safe to proceed



3

7.2 Stage 1 FEPs

Table: FEPs Ranked by Maximum/Average Risk Scores

Rank	FEP	Maximum Risk	Average Risk
1	Toxic geologic components (metals)	20	7.2
2	Contamination of groundwater by CO_2	20	5.8
3	Undetected features	16	7.1
4	Human activities in the surface environment: on site	16	7.5
5	Exogenous economics - supply prices	15	9.8
6	Near-surface aquifers and surface water bodies	15	5.6
7	Accidents and unplanned events: Project	15	5.8
8	Community characteristics	15	5.0
9	Legal/regulatory: property rights and trespass	12	6.9
10	Fractures and faults	12	7.8
11	Schedule and planning	12	6.7
12	Reservoir pore architecture	12	6.2
13	Reservoir geometry	12	5.6
14	Actions and reactions - local community	12	6.6
15	CO_2 solubility and aqueous speciation	12	8.0
16	Seismicity (Induced earthquakes)	12	6.1
17	Actions and reactions - national/international Special Interest Groups (SIGs) and NGOs	12	5.5
18	Legal/regulatory: construction, discharge, and other operations permits	12	5.0
19	Land and water use	12	4.4
20	Compressor procurement	12	7.3
21	Compression facility construction	12	7.4
22	Legal/regulatory: UIC permit	12	7.5
23	Seal failure (in wells)	12	6.9
24	Transportation and logistics	12	3.9
25	Legal and regulatory framework	12	4.5
26	CO_2 injectate quantity and rate	12	4.2
27	Lithology	12	5.3
28	Pre-closure monitoring of storage	12	4.0
29	Actions and reactions - SIGs - local and regional	12	4.6
30	Drilling and Completion activities: Project	12	5.1
31	CO_2 — Health Effects	12	4.0
32	Support from Government — technical basis	12	4.7
33	Support from Government — political basis	12	4.7
34	Support from MGSC partners	12	4.8
35	Technological developments	10	3.7
36	CO_2 phase behavior	10	3.9
37	Shallow Gas/Drift Gas	10	5.0
38	Mineral dissolution — reservoir	10	5.7
39	Mineral dissolution –- borehole	10	5.5
40	Blowouts	10	4.8
41	Hydrogeology	10	3.9

Rank	FEP	Maximum Risk	Average Risk
42	Asphyxiation effects	10	3.9
43	Undefined specification	9	6.0
44	Accidents and unplanned events: External	9	5.4
45	CO_2 release to the atmosphere	9	5.1
46	Mechanical processes and conditions	9	6.8
47	Thermal effects on the injection point in the formation	9	5.3
48	Mineral precipitation	9	5.9
49	CO_2 injectate composition - water vapor	9	3.9
50	Subsurface infrastructure	9	4.0
51	Heterogeneity in reservoir	9	5.5
52	Seal - Geologic, Primary (caprock)	9	5.8
53	Displacement of formation fluid (capillarity)	9	5.3
54	Seismicity (non-project-related earthquakes)	9	3.4
55	Meteorology, weather	9	5.1
56	Heterogeneity of overlying aquifers	9	5.7
57	Monitoring or Verification Wells	9	4.5
58	Overpressurization	9	5.0
59	Quality control	9	4.1
60	Formation damage	9	5.4
61	Well lining and completion	9	5.3
62	CO_2 injectate composition - contaminants	9	3.9
63	Pressure effects on caprock	9	5.4
64	Sealing and closure of boreholes	9	5.3
65	System performance	9	4.8
66	Construction and Operations Activities	9	4.2
67	Soils and sediments	9	5.0
68	Stress and mechanical properties	9	5.5
69	CO_2 supply: interruptions of ethanol production/general ADM plant	9	4.0
70	Formation pressure	9	4.3
71	Construction and Operations Activities other than drilling and well completion	9	4.1
72	Geographic location	9	2.8
73	CO_2 supply: interruptions of compression-dehydration facility	9	3.8
74	Pressure effects on reservoir fluids (other than brine)	9	3.7
75	Mineral dissolution – caprock	8	4.7
76	Model and data issues	8	4.3
77	Lithification and diagenesis	8	4.3
78	Unplanned CO_2 release to the atmosphere	8	4.5
79	Post-project monitoring of storage	8	4.2
80	Lake Nyos effect	8	4.2
81	Security concerns	6	3.6
82	CO_2 release processes	6	3.1
83	Hydrologic and hydrogeologic changes (induced)	6	4.0
84	Changes in groundwater geochemistry (not caused by project)	6	2.6
85	Unconformities	6	3.8
86	Terrestrial flora and fauna	6	3.3
87	Data acquisition activities at well	6	4.4

Rank	FEP	Maximum Risk	Average Risk
88	Tanker transport	6	3.0
89	Data acquisition activities away from well	6	3.1
90	Well records	6	3.8
91	Seal - Geologic, additional	6	4.0
92	Records and markers	6	2.4
93	Human activities in the surface environment: off site	6	3.1
94	Digester tanks	6	3.4
95	Explosions and crashes	6	3.3
96	Orphan wells	6	3.8
97	Desiccation of clay	6	3.8
98	Gas chemistry	6	3.3
99	Water-mediated transport	6	3.0
100	Drilling activities: Non-project	6	2.8
101	Post-project administrative control	6	3.0
102	Pre-closure administrative control	6	2.3
103	Social and institutional developments	6	2.2
104	Meteorite impact	5	4.0
105	Toxic geologic components (gases)	5	3.3
106	Buildings	5	2.8
107	Displacement of saline formation fluid (long-distance effects)	5	2.5
108	Subsidence or uplift	5	2.7
109	Unit conversions (English/Metric, etc.)	4	2.4
110	Erosion and deposition	4	2.1
111	Vertical geothermal gradient	4	2.3
112	Exogenous economics — general	4	2.2
113	Mining and other underground activities	4	2.4
114	Hydrological regime and water balance	4	2.0
115	Topography and morphology	4	1.7
116	Exogenous economics - CCS-specific	4	1.8
117	Future human actions	4	1.8
118	Buoyancy-driven flow	4	1.8
119	Impacts on soils and sediments	3	2.2

7.3 Stage 1 Scenarios

To facilitate review by responsible persons within appropriate professional disciplines, scenarios were organized into 29 groups having ID numbers and labels as shown in the first table below. The second table lists the grouped individual scenarios.

Scenario Groups

OE01	QHSE Plan: Subsurface And Seismic Operations
OE02	QHSE Plan: Surface And Injection Operations
OE03	Site Security Plan
OE04	Drilling Plans, Deep Wells
OE05	Completion Plan: In-Zone Wells
OE06	Completion Plan: Geophone Wells
OE07	Completion Plan: Shallow Groundwater Monitoring Wells
OE08	Co_2-Resistant Well Construction -- In-Zone Wells
OE09	Injection Operations Envelope
OE10	Injection Operations and Shut-in Plan
OE11	Equipment Sparing Plan
OE12	Operational Monitoring Plan
PM01	Overall Risk Management
PM02	Permits
PM03	Data Management Plan
PM04	Budget
PM05	Staff And Coordination
PM08	Communications Policy
PM12	Injectivity Backup Plan
PM13	Co_2 Movement Contingency Plan
PM14	Goal Triage
PM15	Communications Plan
SR01	Site Selection and AOR
SR02	USDW Baseline Chemistry
SR03	Reservoir Characterization Plan: In-Zone Wells
SR06	MVA Plan
SR08	Surface Seismic
SR09	Static Earth Model
SR10	Simulation Plan

In Stage 1, some scenarios were deemed critical to multiple areas and were managed as members of multiple scenario groups. Risk treatments were tracked against specific scenarios through the Scenario ID number unique to each scenario.

Initial scenario risk ranking shown in last column; scenarios lacking rank number were generated after the FEPs process.

Individual Scenarios

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
OE01	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
OE01	#Hs06	Rupture in CO_2 pipeline or wellhead allows rapid gas escape, creating mechanical and temperature hazards.	4
OE01	#Hs05	Mechanically caused rupture (e.g., equipment collision) in CO_2 pipeline or wellhead creates toxic atmosphere or Immediate Danger to Life or Health (IDLH) low-oxygen condition.	4
OE01	#Hs11	Moving vehicle strikes person or damages equipment; traffic hazards onsite or offsite.	4
OE01	#Hs12	Numerous common industrial, drilling, and driving hazards, typically addressed by existing Quality, Health, Safety, Environment (QHSE) practices.	7
OE01	#Hs14	Personal weather hazards: Cold, heat, wind, lightning, etc.	7
OE01	#Op03	Unfamiliar workflows result in delays impacting budget. Tensions develop among parties leading to a loss of focus on HSE, resulting in an accident or unplanned rig event.	11
OE01	#Hs10	Unplanned CO_2 release in confined space could cause health hazard to employees working on compressor and other surface facilities.	31
OE01	#Hs09	Chemical upset condition ruptures vessel, pipeline, or wellhead, releasing CO_2 at IDLH rate.	78
OE02	#Hs12	Numerous common industrial, drilling, and driving hazards, typically addressed by existing QHSE practices.	7
OE02	#Op03	Unfamiliar workflows result in delays impacting budget. Tensions develop among parties leading to a loss of focus on HSE, resulting in an accident or unplanned rig event.	11
OE02	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
OE03	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
OE03	#Hs06	Rupture in CO_2 pipeline or wellhead allows rapid gas escape, creating mechanical and temperature hazards.	4

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
OE03	#Hs05	Mechanically caused rupture (e.g., equipment collision) in CO_2 pipeline or wellhead creates toxic atmosphere or IDLH low-oxygen condition.	4
OE03	#Hs11	Moving vehicle strikes person or damages equipment; traffic hazards onsite or offsite.	4
OE03	#Hs09	Chemical upset condition ruptures vessel, pipeline, or wellhead, releasing CO_2 at IDLH rate.	78
OE04	#Hs02	CO_2 moves via wellbore or fault directly into Underground Source of Drinking Water (USDW).	1
OE04	#Hs03	CO_2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW.	1
OE04	#Hs04	CO_2 acidifies groundwater which mobilizes metals already present within the USDW lithology.	1
OE04	#Hs08	Very large mass of CO_2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO_2 concentration in a subgrade space or topographic low.	10
OE04	#Rs06	Lack of coordination causes loss of measurement opportunity or loss of acquired data.	11
OE04	#Mg02	The second verification well is drilled after injection has begun. While drilling the well, CO_2 is encountered above the Eau Claire, resulting in a well-control issue. Project is suspended until leakage path is identified.	40
OE04	#Mg03	While performing yearly monitoring operations, equipment failure or human error allows CO_2 to escape uncontrolled from the well for several hours or days.	45
OE05	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
OE05	#Hs03	CO_2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW.	1
OE05	#Hs04	CO_2 acidifies groundwater which mobilizes metals already present within the USDW lithology.	1
OE05	#Hs08	Very large mass of CO_2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO_2 concentration in a subgrade space or topographic low.	10
OE05	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
OE06	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
OE06	#Hs03	CO_2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW.	1
OE06	#Hs04	CO_2 acidifies groundwater which mobilizes metals already present within the USDW lithology.	1

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
OE06	#Hs08	Very large mass of CO_2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO_2 concentration in a subgrade space or topographic low.	10
OE07	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
OE07	#Hs03	CO_2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW.	1
OE07	#Hs04	CO_2 acidifies groundwater which mobilizes metals already present within the USDW lithology.	1
OE07	#Hs08	Very large mass of CO_2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO_2 concentration in a subgrade space or topographic low.	10
OE08	#Ci20	Action causes failure to comply with permit.	1
OE08	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
OE08	#Hs03	CO_2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW.	1
OE08	#Hs04	CO_2 acidifies groundwater which mobilizes metals already present within the USDW lithology.	1
OE08	#Hs08	Very large mass of CO_2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO_2 concentration in a subgrade space or topographic low.	10
OE08	#Op04	Cement dissolution results in very small amounts of CO_2 gas present in intermediate/injection string annulus after two years of injection, causing remedial work to be performed to finish project.	15
OE08	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
OE09	#Ci06	An unexpected situation arises that is not technically impactful, but the appearance of unpreparedness spurs public concern.	1
OE09	#Ci20	Action causes failure to comply with permit.	1
OE09	#Zz01	An unexpected situation arises for which prep is inadequate, thus impact occurs.	1
OE09	#Ci09	Public knowledge of significant planning error causes loss of public confidence in ability to conduct CCS.	1
OE09	#Hs05	Mechanically caused rupture (e.g., equipment collision) in CO_2 pipeline or wellhead creates toxic atmosphere or IDLH low-oxygen condition.	4
OE09	#Hs06	Rupture in CO_2 pipeline or wellhead allows rapid gas escape, creating mechanical and temperature hazards.	4
OE09	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
OE09	#Fn03	Cost implications of any accident, including emergency response, medical, equipment damage, delay(s).	4

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
OE09	#Fn01	Price escalation for materials or consumables results in budget overrun; not all project goals can be met.	5
OE09	#Ci07	Prominent accident (especially involving CO_2 release) raises questions about general safety of CCS -- even if the accident is ordinary industrial or drilling, not related to CCS specifically.	7
OE09	#Hs14	Personal weather hazards: Cold, heat, wind, lightning, etc.	7
OE09	#Op02	Severe weather necessitates suspension of operations for several days, due to inability of personnel to reach site and/or to work safely.	7
OE09	#Fn05	Low permeability forces redesign of CO_2 delivery system, incurring direct costs and delay(s).	12
OE09	#Rs10	Procurement delay shortens the time available for tests using a critical piece of equipment, causing loss of baseline data for the equipment, the site, or both.	20
OE09	#Mg07	CO_2 migrates across well seal (cement/casing) at a rate high enough that injection is suspended, preventing achievement of project goals.	23
OE09	#Rs11	Breakdown of sensor or tool; replacement or repair delay causes irretrievable information loss.	24
OE09	#Ci14	An accident by any member of the team reflects badly upon the project, and/or upon the image of another consortium member.	30
OE09	#Hs10	Unplanned CO_2 release in confined space could cause health hazard to employees working on compressor and other surface facilities.	31
OE09	#Ci23	Increased national urgency to execute CCS quickly forces injection to be implemented with less baseline data or less monitoring than planned.	33
OE09	#Ci24	Increased national urgency to execute CCS cheaply forces injection to be implemented with less baseline data or less monitoring than planned.	33
OE09	#Op13	Manufacturing of key mechanical components ceases, and insufficient spares have been stocked to meet repair and/or maintenance requirements for the long-term, incurring extra cost for custom manufacture.	35
OE09	#Op14	Information technology changes and archival data are lost due to inability to read older media or formats.	35
OE09	#Op11	Equipment breakdown causes repair expense and delay.	43
OE09	#Mg03	While performing yearly monitoring operations, equipment failure or human error allows CO_2 to escape uncontrolled from the well for several hours or days.	45
OE09	#Rs13	Weather conditions prevent using a monitoring method or retrieving data at a critical point in time.	55
OE09	#Fn16	Poor execution of project activity necessitates project activity to be repeated.	59
OE09	#Op15	Intermittent or poorly controlled injection (temperature, pressure, flow rate, composition) causes rapid degradation of downhole components through chemical or mechanical means.	65
OE09	#Hs09	Chemical upset condition ruptures vessel, pipeline, or wellhead, releasing CO_2 at IDLH rate.	78

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
OE09	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
OE10	#Ci06	An unexpected situation arises that is not technically impactful, but the appearance of unpreparedness spurs public concern.	1
OE10	#Hs01	CO_2 laterally displaces Mt. Simon brine with dissolved metals, moving brine to where Mt. Simon had freshwater/is a USDW.	1
OE10	#Zz01	An unexpected situation arises for which prep is inadequate, thus impact occurs.	1
OE10	#Mg01	CO_2 migrates to undiscovered, unsealed orphan well that provides a leakage path.	3
OE10	#Hs05	Mechanically caused rupture (e.g., equipment collision) in CO_2 pipeline or wellhead creates toxic atmosphere or IDLH low-oxygen condition.	4
OE10	#Hs06	Rupture in CO_2 pipeline or wellhead allows rapid gas escape, creating mechanical and temperature hazards.	4
OE10	#Hs11	Moving vehicle strikes person or damages equipment; traffic hazards onsite or offsite.	4
OE10	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
OE10	#Fn03	Cost implications of any accident, including emergency response, medical, equipment damage, delay(s).	4
OE10	#Op02	Severe weather necessitates suspension of operations for several days, due to inability of personnel to reach site and/or to work safely.	7
OE10	#Mg06	Subsurface condition causes plume migration beneath sensitive area or causes unexpected long-distance migration.	9
OE10	#Fn04	Lack of coordination of activities causes resource or timing conflict, causing delay.	11
OE10	#Fn10	Unexpected failure requires remediation that is not budgeted.	23
OE10	#Mg07	CO_2 migrates across well seal (cement/casing) at a rate high enough that injection is suspended, preventing achievement of project goals.	23
OE10	#Hs10	Unplanned CO_2 release in confined space could cause health hazard to employees working on compressor and other surface facilities.	31
OE10	#Ci23	Increased national urgency to execute CCS quickly forces injection to be implemented with less baseline data or less monitoring than planned.	33
OE10	#Ci24	Increased national urgency to execute CCS cheaply forces injection to be implemented with less baseline data or less monitoring than planned.	33
OE10	#Mg02	The second verification well is drilled after injection has begun. While drilling the well, CO_2 is encountered above the Eau Claire, resulting in a well-control issue. Project is suspended until leakage path is identified.	40
OE10	#Mg03	While performing yearly monitoring operations, equipment failure or human error allows CO_2 to escape uncontrolled from the well for several hours or days.	45

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
OE10	#Sf03	Major earthquake causes unexpected damage.	54
OE10	#Fn16	Poor execution of project activity necessitates project activity to be repeated.	59
OE10	#Hs09	Chemical upset condition ruptures vessel, pipeline, or wellhead, releasing CO_2 at IDLH rate.	78
OE10	#Op06	Numerous minor traffic accidents occur at or around the site, resulting in ADM losing some confidence in project management and requiring a safety shutdown to hold additional training.	
OE10	#Op11	Equipment breakdown causes repair expense and delay(s).	
OE10	#Op13	Manufacturing of key mechanical components ceases, and insufficient spares have been stocked to meet repair and/or maintenance requirements for the long-term, incurring extra cost for custom manufacture.	
OE10	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
OE11	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
OE11	#Rs11	Breakdown of sensor or tool; replacement or repair delay causes irretrievable information loss.	24
OE11	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
OE12	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
OE12	#Mg07	CO_2 migrates across well seal (cement/casing) at a rate high enough that injection is suspended, preventing achievement of project goals.	23
OE12	#Op15	Intermittent or poorly controlled injection (temperature, pressure, flow rate, composition) causes rapid degradation of downhole components through chemical or mechanical means.	65
OE12	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
PM01	#Fn15	Claim of project-caused damage to geologic resource (water, gas, oil, coal), agriculture, or buildings must be defended. Successful claim adds remedial cost or damage payment.	1
PM01	#Zz01	An unexpected situation arises for which prep is inadequate, thus impact occurs.	1
PM02	#Ci21	Action causes public opposition which results in permit or project delay.	1
PM02	#Ci19	Action causes permit delay.	1
PM02	#Ci20	Action causes failure to comply with permit.	1
PM02	#Op05	Lack of local operational experience results in not obtaining a required permit. For biggest media impact, regulators wait to intercede until just before activities begin, resulting in lost days.	18
PM03	#Ci20	Action causes failure to comply with permit.	1
PM03	#Ci17	Different reservoir models show different result; noticed first primarily in the technical community, but later by the media or the general public, resulting in loss of confidence in technical competence.	1

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
PM03	#Op12	Injection well drilling encounters borehole stability problems requiring changes to casing and/or mud program, resulting in lost days, increased drilling cost, and/or lost measurement opportunities.	30
PM03	#Op14	Information technology changes, and archival data are lost due to inability to read older media or formats.	35
PM04	#Ci21	Action causes public opposition which results in permit or project delay.	1
PM04	#Ci20	Action causes failure to comply with permit.	1
PM04	#Mg01	CO_2 migrates to undiscovered, unsealed orphan well that provides a leakage path.	3
PM04	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
PM04	#Ci15	Cost overruns and failure to meet all goals result in doubt cast on ability of MGSC and Schlumberger Carbon Services to manage such projects.	5
PM04	#Rs01	Escalation in cost of research consumables or services reduces the remaining research budget, thus not all goals can be met.	5
PM04	#Rs02	Overall research budget is reduced regardless of cause, thus not all goals can be met.	5
PM04	#Rs12	Incorrect or inadequate physical process models incorrectly predict plume fate, causing incorrectly targeted data acquisition.	27
PM05	#Ci21	Action causes public opposition which results in permit or project delay.	1
PM05	#Ci19	Action causes permit delay.	1
PM05	#Fn15	Claim of project-caused damage to geologic resource (water, gas, oil, coal), agriculture, or buildings must be legally defended. If claim is successful, potential additional remedial cost or damage payment.	1
PM05	#Fn12	General: change in project scope requires additional funding.	3
PM05	#Rs01	Escalation in cost of research consumables or services reduces the remaining research budget, thus not all goals can be met.	5
PM05	#Rs06	Lack of coordination causes loss of measurement opportunity or loss of acquired data.	11
PM05	#Fn04	Lack of coordination of activities causes resource or timing conflict, causing delay.	11
PM05	#Op03	Unfamiliar workflows result in delays impacting budget. Tensions develop among parties leading to a loss of focus on HSE, resulting in an accident or unplanned rig event.	11
PM05	#Fn08	A permit is modified by government agency to require increased data access and approval by NGOs and/or public; this causes delays, leading to price escalation.	17
PM05	#Rs10	Procurement delay shortens the time available for tests using a critical piece of equipment, causing loss of baseline data about either this equipment or the site.	20

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
PM05	#Fn07	Procurement delay for critical equipment causes increased costs due to rescheduling, and/or overall project delay.	20
PM08	#Ci04	Unreleased project monitoring data that is of technical concern reaches the public and spurs public concern.	1
PM08	#Ci17	Different reservoir models show different result; noticed first primarily in the technical community, but later by the media or the general public, resulting in loss of confidence in technical competence.	1
PM08	#Fn15	Claim of project-caused damage to geologic resource (water, gas, oil, coal), agriculture, or buildings must be legally defended. If claim is successful, potential additional remedial cost or damage payment.	1
PM08	#Mg01	CO_2 migrates to undiscovered, unsealed orphan well that provides a leakage path.	3
PM08	#Fn04	Lack of coordination of activities causes resource or timing conflict, causing delay.	11
PM08	#Op03	Unfamiliar workflows result in delays impacting budget. Tensions develop among parties leading to a loss of focus on HSE, resulting in an accident or unplanned rig event.	11
PM08	#Rs06	Lack of coordination causes loss of measurement opportunity or loss of acquired data.	11
PM08	#Op04	Cement dissolution results in very small amounts of CO_2 gas present in intermediate/injection string annulus after two years of injection, causing remedial work to be performed to finish project.	15
PM08	#Fn08	A permit is modified by government agency to require increased data access and approval by NGOs and/or public; this causes delays, leading to price escalation.	17
PM08	#Op05	Lack of local operational experience results in not obtaining a required permit. For biggest media impact, regulators wait to intercede until just before activities begin, resulting in lost days.	18
PM08	#Rs09	Public opposition based on feared impact to land use (e.g., farming) impedes using monitoring method.	19
PM08	#Mg07	CO_2 migrates across well seal (cement/casing) at a rate high enough that injection is suspended, preventing achievement of project goals.	23
PM08	#Ci23	Increased national urgency to execute CCS quickly forces injection to be implemented with less baseline data or less monitoring than planned.	33
PM08	#Ci24	Increased national urgency to execute CCS cheaply forces injection to be implemented with less baseline data or less monitoring than planned.	33
PM08	#Op14	Information technology changes and archival data are lost due to inability to read older media or formats.	35
PM12	#Rs14	Overestimation of fracture gradient. Mt. Simon is supposed to have a 1 psi/ft fracture gradient and all injection modeling is based upon this. If testing shows that fracture gradient is actually much lower (0.6), this would have a major impact on injection rates as well as causing the IEPA to rethink injection permit.	12

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
PM12	#Fn05	Low permeability forces redesign of CO_2 delivery system, incurring direct cost and delay.	12
PM12	#Rs07	Failure to find adequate porosity/permeability zones prevents injecting at desired rate, prevents conducting desired tests, and/or could cause project cancellation.	12
PM12	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
PM13	#Ci20	Action causes failure to comply with permit.	1
PM13	#Mg01	CO_2 migrates to undiscovered, unsealed orphan well that provides a leakage path.	3
PM13	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
PM13	#Hs06	Rupture in CO_2 pipeline or wellhead allows rapid gas escape, creating mechanical and temperature hazards.	4
PM13	#Hs05	Mechanically caused rupture (e.g., equipment collision) in CO_2 pipeline or wellhead creates toxic atmosphere or IDLH low-oxygen condition.	4
PM13	#Hs11	Moving vehicle strikes person or damages equipment; traffic hazards onsite or offsite.	4
PM13	#Hs09	Chemical upset condition ruptures vessel, pipeline, or wellhead, releasing CO_2 at IDLH rate.	78
PM14	#Ci21	Action causes public opposition which results in permit or project delay.	1
PM14	#Ci19	Action causes permit delay.	1
PM14	#Ci09	Public knowledge of significant planning error causes loss of public confidence in ability to conduct CCS.	1
PM14	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
PM14	#Fn01	Price escalation for materials or consumables results in budget overrun; not all project goals can be met.	5
PM14	#Fn02	Price escalation for labor, services, lease/royalty payments, or insurance results in budget overrun; not all goals can be met.	5
PM14	#Ci15	Cost overruns and failure to meet all goals result in doubt cast on ability of MGSC and Schlumberger Carbon Services to manage such projects.	5
PM14	#Rs02	Overall research budget is reduced regardless of cause, thus not all goals can be met.	5
PM14	#Rs04	Subsurface condition impairs seismic imaging, groundwater sampling, logging, or other data acquisition, impairing ability to track plume.	10
PM14	#Op05	Lack of local operational experience results in not obtaining a required permit. For biggest media impact, regulators wait to intercede until just before activities begin, resulting in lost days.	18
PM14	#Rs10	Procurement delay shortens the time available for tests using a critical piece of equipment, causing loss of baseline data about either this equipment or the site.	20
PM14	#Fn07	Procurement delay for critical equipment causes increased costs due to rescheduling, and/or overall project delay.	20

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
PM14	#Mg07	CO_2 migrates across well seal (cement/casing) at a rate high enough that injection is suspended, preventing achievement of project goals.	23
PM14	#Op12	Injection well drilling encounters borehole stability problems requiring changes to casing and/or mud program, resulting in lost days, increased drilling cost, and/or lost measurement opportunities.	30
PM14	#Ci24	Increased national urgency to execute CCS cheaply forces injection to be implemented with less baseline data or less monitoring than planned.	33
PM14	#Ci23	Increased national urgency to execute CCS quickly forces injection to be implemented with less baseline data or less monitoring than planned.	33
PM14	#Op11	Equipment breakdown causes repair expense and delay.	43
PM14	#Op16	Incorrect or inadequate physical process models incorrectly predict response of engineered components, leading to unexpected failure.	
PM15	#Ci01	Public concern about water mobilizes opposition that impedes project activities, arising without a specific triggering event.	1
PM15	#Ci04	Unreleased project monitoring data that is of technical concern reaches the public and spurs public concern.	1
PM15	#Ci05	Non-project-related groundwater change is ascribed to project, spurring public concern.	1
PM15	#Ci06	An unexpected situation arises that is not technically impactful, but the appearance of unpreparedness spurs public concern.	1
PM15	#Ci09	Public knowledge of significant planning error causes loss of public confidence in ability to conduct CCS.	1
PM15	#Ci10	Public fears of CO_2 injection or monitoring, though technically not well-founded, create opposition that impedes operations.	1
PM15	#Ci17	Different reservoir models show different result; noticed first primarily in the technical community, but later by the media or the general public, resulting in loss of confidence in technical competence.	1
PM15	#Ci21	Action causes public opposition which results in permit or project delay.	1
PM15	#Fn15	Claim of project-caused damage to geologic resource (water, gas, oil, coal), agriculture, or buildings must be legally defended. If claim is successful, potential additional remedial cost or damage payment.	1
PM15	#Mg01	CO_2 migrates to undiscovered, unsealed orphan well that provides a leakage path.	3
PM15	#Ci15	Cost overruns and failure to meet all goals result in doubt cast on ability of MGSC and Schlumberger Carbon Services to manage such projects.	5
PM15	#Op04	Cement dissolution results in very small amounts of CO_2 gas present in intermediate/injection string annulus after two years of injection, causing remedial work to be performed to finish project.	15
PM15	#Fn08	A permit is modified by government agency to require increased data access and approval by NGOs and/or public; this causes delays, leading to price escalation.	17

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
PM15	#Op05	Lack of local operational experience results in not obtaining a required permit. For biggest media impact, regulators wait to intercede until just before activities begin, resulting in lost days.	18
PM15	#Rs09	Public opposition based on feared impact to land use (e.g., farming) impedes using monitoring method.	19
PM15	#Mg07	CO_2 migrates across well seal (cement/casing) at a rate high enough that injection is suspended, preventing achievement of project goals.	23
SR01	#Ci19	Action causes permit delay.	1
SR01	#Ci20	Action causes failure to comply with permit.	1
SR01	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
SR01	#Hs01	CO_2 laterally displaces Mt. Simon brine with dissolved metals, moving brine to where Mt. Simon had freshwater/is a USDW.	1
SR01	#Mg01	CO_2 migrates to undiscovered, unsealed orphan well that provides a leakage path.	3
SR02	#Ci20	Action causes failure to comply with permit.	1
SR02	#Ci06	An unexpected situation arises that is not technically impactful, but the appearance of unpreparedness spurs public concern.	1
SR02	#Ci05	Non-project-related groundwater change is ascribed to project, spurring public concern.	1
SR03	#Ci20	Action causes failure to comply with permit.	1
SR03	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
SR03	#Hs03	CO_2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW.	1
SR03	#Hs04	CO_2 acidifies groundwater which mobilizes metals already present within the USDW lithology.	1
SR03	#Hs08	Very large mass of CO_2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO_2 concentration in a subgrade space or topographic low.	10
SR03	#Rs07	Failure to find adequate porosity/permeability zones prevents injecting at desired rate, prevents conducting desired tests, could cause project cancellation.	12
SR06	#Ci19	Action causes permit delay.	1
SR06	#Mg05	CO_2 exits storage reservoir and crosses caprock via fault or fracture.	1
SR06	#Ci20	Action causes failure to comply with permit.	1
SR06	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
SR06	#Ci06	An unexpected situation arises that is not technically impactful, but the appearance of unpreparedness spurs public concern.	1
SR06	#Fn15	Claim of project-caused damage to geologic resource (water, gas, oil, coal), agriculture, or buildings must be legally defended. If claim is successful, potential additional remedial cost or damage payment.	1
SR06	#Hs01	CO_2 laterally displaces Mt. Simon brine with dissolved metals, moving brine to where Mt. Simon had freshwater/is a USDW.	1

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
SR06	#Rs03	Accident damages custom-manufactured equipment, sensor, tool; replacement or repair delay causes irretrievable information loss during an injection phase.	4
SR06	#Rs05	Subsurface condition causes plume migration into area where monitoring ability is impaired.	6
SR06	#Hs15	Groundwater with high CO_2, metals, toxic gas, or acid content moves into shallow aquifer or surface water body.	6
SR06	#Mg06	Subsurface condition causes plume migration beneath sensitive area or causes unexpected long-distance migration.	9
SR06	#Rs06	Lack of coordination causes loss of measurement opportunity or data.	11
SR06	#Rs11	Breakdown of sensor or tool; replacement or repair delay causes irretrievable information loss.	24
SR08	#Mg05	CO_2 exits storage reservoir and crosses caprock via fault or fracture.	1
SR08	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
SR08	#Hs01	CO_2 laterally displaces Mt. Simon brine with dissolved metals, moving brine to where Mt. Simon had freshwater/is a USDW.	1
SR09	#Mg05	CO_2 exits storage reservoir and crosses caprock via fault or fracture.	1
SR09	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
SR09	#Ci17	Different reservoir models show different result; noticed first primarily in the technical community, but later by the media or the general public, resulting in loss of confidence in technical competence.	1
SR09	#Hs01	CO_2 laterally displaces Mt. Simon brine with dissolved metals, moving brine to where Mt. Simon had freshwater/is a USDW.	1
SR09	#Ci16	Model results that become public suggest plume leaving site. Adjacent land owners decide to test laws of pore space ownership resulting in lengthy court cases. Project on hold for lengthy period of time.	8
SR09	#Mg06	Subsurface condition causes plume migration beneath sensitive area or causes unexpected long-distance migration.	9
SR09	#Fn06	Plume travels off property controlled by project partner, incurring legal action or payment demand by adjacent property owner.	9
SR09	#Hs08	Very large mass of CO_2 continuously migrates via faults or fractures across multiple seals to soil or atmosphere, creating high CO_2 concentration in a subgrade space or topographic low.	10
SR09	#Ci12	Small earthquakes occur and though they are scientifically shown to be unrelated to project, public/media concern forces project to devote time to debunking rumors.	10
SR09	#Op08	Injection well drilling encounters unexpected high pressures requiring well-killing operation resulting in lost days, increased drilling cost, and/or lost measurement opportunities.	30
SR09	#Sf04	Drilling or mining for unrelated purpose encounters unexpected high pressure fluid, damaging equipment and/or resource.	74
SR10	#Mg05	CO_2 exits storage reservoir and crosses caprock via fault or fracture.	1
SR10	#Ci20	Action causes failure to comply with permit.	1

Group	Scenario ID	Scenario	Highest Rank of Related FEPs
SR10	#Hs02	CO_2 moves via wellbore or fault directly into USDW.	1
SR10	#Hs03	CO_2-acidified brine moves via wellbore or fault, dissolving metals along the way, into USDW.	1
SR10	#Ci05	Non-project-related groundwater change is ascribed to project; public concern.	1
SR10	#Ci17	Different reservoir models show different result; noticed first primarily in the technical community, but later by the media or the general public, resulting in loss of confidence in technical competence.	1
SR10	#Hs04	CO_2 acidifies groundwater which mobilizes metals already present within the USDW lithology.	1
SR10	#Hs01	CO_2 laterally displaces Mt. Simon brine with dissolved metals, moving brine to where Mt. Simon had freshwater/is a USDW.	1
SR10	#Ci16	Model results that become public suggest plume leaving site. Adjacent land owners decide to test laws of pore space ownership resulting in lengthy court cases. Project on hold for lengthy period of time.	8
SR10	#Mg06	Subsurface condition causes plume migration beneath sensitive area or causes unexpected long-distance migration.	9
SR10	#Fn06	Plume travels off property controlled by project partner, incurring legal action or payment demand by adjacent property owner.	9
SR10	#Rs07	Failure to find adequate porosity/permeability zones prevents injecting at desired rate, prevents conducting desired tests, could cause project cancellation.	12
SR10	#Rs12	Incorrect or inadequate physical process models incorrectly predict plume fate, causing incorrectly targeted data acquisition.	27
SR10	#Op15	Intermittent or poorly controlled injection (temperature, pressure, flow rate, composition) causes rapid degradation of downhole components through chemical or mechanical means.	65

7.4 Stage 1 Risk Treatments

A single scenario may require multiple risk treatments, and a single treatment may reduce risk for multiple scenarios; one-to-one relationships are not typical. For a long-term project, the effort to develop a relational database to manage these complex many-to-many relationships would be justified. For IBDP, the complexity was simplified by topically grouping risk treatments.

Note: Risk treatments generally correspond to the scenarios in the scenario group of the same name. Risk treatments were given unique ID numbers for tracking the completion of treatment tasks.

Risk Response Action Groups and Risk Treatments

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
OE01	QHSE PLAN: Subsurface and Seismic Operations	00001	Prepare a detailed QHSE Site Plan to cover construction, downhole, drilling/completion, and seismic activities not involving CO_2 that includes at least the following:
OE01		00002	Define minimum initial and recurrent training requirements for workers and visitors, confined-space activities, weather hazards, motor vehicle safety, on-site and off-site work, and all other QHSE issues that might arise pertinent to these activities.
OE01		00003	Ensure that no one works at site before completing the provided safety training.
OE01		00007	Ensure that Journey Management Plan (JMP) and Site Traffic Plan (STP) are written, communicated, and trained for.
OE01		00008	Ensure that performance per JMP and STP is tracked.
OE01		00010	Ensure the plan includes a Bridging Document that covers the specific organizations involved in these activities.
OE02	QHSE PLAN: Surface and Injection Operations	00004	During project phases and near locations where high-pressure CO_2 is present, identify hazards specific to its presence, and incorporate safety procedures into Surface QHSE Site plan.
OE02		00006	Ensure that procedures, training, and available equipment support the prevention of anyone working in a confined space without proper ventilation and active oversight.
OE02		00011	Prepare detailed Surface QHSE Site Plan to cover surface and injection operations.
OE02		00012	Ensure that Surface QHSE Site Plan covers minimum initial and recurrent training requirements for workers and visitors, weather hazards, motor vehicle safety, on-site and off-site work, all areas of site, site office, and injection-well end of pipeline, and safety equipment to be stored at site office.
OE02		00013	Perform risk-reduction measures specific to scenarios of high-flow CO_2 release, especially in proximity to confined spaces, including the injection-well cellar and semi-confined spaces such as berm-bounded areas.
OE02		00014	Ensure that no one works at site before completing the provided safety training.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
OE02		00015	Ensure the presence and continuous functionality of an alarm system to warn persons on- and near-site in the event of significant above-background CO_2 in air, with planned project-specific alert levels and response protocols, and with monitoring locations to include EC tower, well pad, and site office.
OE02	QHSE PLAN: Surface and Injection Operations	00017	Ensure that JMP and STP are written, communicated, and trained for, and that performance per JMP and STP is tracked.
OE02		00018	Prepare a crisis call list and review it monthly for currency.
OE02		00019	Ensure that plan includes a Bridging Document that covers the specific organizations involved in these activities.
OE02		00020	Include HAZOP exercises as necessary.
OE02		01008	Review Subsurface QHSE Plan, and ensure that any project QHSE aspect that is not specifically addressed by the Subsurface QHSE Plan is addressed by the Surface QHSE Site Plan.
OE03	SITE SECURITY PLAN	00021	Develop site security plan that includes active 24/7 security, fencing, signage, warning devices, sensors, alarms, physical barriers to prevent rupture from accidental impact (e.g., by truck) upon key components including wellhead, pipeline.
OE04	DRILLING PLANS, DEEP WELLS	00022	Ensure that a detailed drilling plan is in place covering at least these elements:
OE04		00023	Construction plan (suited as appropriate to conditions pre-injection, during injection, or after well control operations);
OE04		00024	Drill site security;
OE04		00025	Wellbore security;
OE04		00026	Operations to reestablish normal mud balance and/or circulating conditions (regaining control within several hours at most);
OE04		00028	Equipment and personnel training requirements;
OE04		00029	Drill Well On Paper (DWOP) and Complete Well On Paper (and CWOP) exercises.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
OE05	COMPLETION PLAN: IN-ZONE WELLS	00030	Write a detailed plan for completing any injection or monitoring well, covering at least these elements:
OE05		00031	Inspection and verification of wellbore mechanical integrity, and reconfirmation as necessary if significant time has passed before starting injection;
OE05		00032	Regulatory required specifications; additional logging needs;
OE05		00033	Timetable for ordering required equipment.
OE06	COMPLETION PLAN: GEOPHONE WELLS	00034	Write a detailed plan for completing any injection or monitoring well, covering at least these elements:
OE06		00035	Inspection and verification of wellbore mechanical integrity, and reconfirmation, as necessary, if significant time has passed before starting injection;
OE06		00036	Regulatory required specifications;
OE06		00037	Additional logging needs;
OE06		00038	Timetable for ordering required equipment.
OE07	COMPLETION PLAN: SHALLOW GW MON WELLS	00039	Write a detailed plan for completing any monitoring well, covering at least these elements:
OE07		00040	Inspection and verification of wellbore mechanical integrity;
OE07		00041	Regulatory required specifications;
OE07		00042	Additional logging needs;
OE07		00043	Timetable for ordering required equipment.
OE08	CO_2 - RESISTANT WELL CONSTRUCTION: IN-ZONE WELLS	00044	To counter a potentially corrosive downhole environment that may feature CO_2 -saturated brine:
OE08		00045	Use chrome steel casing and special CO_2-resistant cement;
OE08		00046	Ensure that cement placement and Mechanical Integrity Testing (MIT) are performed to the highest standards.
OE08		00047	Ensure that CO_2-resistant well construction characteristics are accurately and thoroughly represented in the final submitted Completion Report.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
OE09	INJECTION OPERATIONS ENVELOPE	00058	Envelope Maximization: To minimize risk to optimal operating conditions and avoid negative impacts, design all mechanical systems to operate under as wide a range of conditions as may be practically achievable. Further details are specified:
OE09		00059	Include dehydration, purification, compression, heating/cooling, pipeline, tanks, and distribution systems;
OE09		00060	Address CO_2 delivery at the wellhead at a range of pressures, temperatures, purities, and flow rates, up to the maximum range of each individual parameter;
OE09		00061	Address maximum ranges (as allowed by various limitations), including downhole injection pressure, maximum/minimum allowed temperature profile in the well, and corrosion/geochemistry/HSE/legal considerations on the injectate composition;
OE09		00062	Avoid design elements that would operate poorly outside of the optimal design point;
OE09		00063	Characterize the impact on capital and operating expenditures of operational variations within the proposed operating envelope;
OE09		00064	Provide a carbon and energy balance of the injection system operating at design conditions;
OE09		00065	Ensure the conditions of delivery of CO_2 from the separation/transport equipment and the supply of CO_2 to the wells are consistent with the respective components' design specifications;
OE09		00066	Refine design specifications early, ensuring that newly acquired information on the geological environment or the wells' completions is quickly reviewed for potential effect upon delivery-system design, and communicated as needed.
OE09		00067	Envelope Definition: To support ongoing injection operation within the range of expected safe conditions, create a concise statement (supported by clear graphics and charts) of the estimated operating range envelope, containing at least these elements and characteristics.
OE09		00068	Foundation in calculations and simulations based on best current physicochemical understandings;
OE09		00069	Description of the boundaries to the ranges of conditions within which the system can be allowed to vary without significant infrastructure implications and/or without significant subsurface implications;
OE09		00070	Description of the likely implications of operations outside the described safe envelope;
OE09		00071	Justification (from modeling or passive/active control mechanisms) that the delivery/transport/subsurface systems can operate in a stable and safe manner together across the whole operating range envelope;
OE09		00072	Demonstration that the system can be controlled or shut-down safely in case of departure of any sub-system from the operating range envelope;

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
OE09		00073	Demonstration that proposed process monitoring and control equipment are sufficient to define the system state within the operating range envelope, and that unwanted departures can be safely identified and corrected.
OE10	INJECTION OPERATIONS AND SHUT-IN PLAN	00048	Develop a site-specific injection operating plan that includes at least these elements:
OE10		00049	Plan technical basis informed by results from Integrated Asset Modeling;
OE10		00050	Constant injection operations, planned shut-ins, and unplanned shut-ins;
OE10		00051	Protecting operating equipment and adjusting operations to account for weather stresses (e.g., output temperature) and expected inflow variations;
OE10		00052	Plan, procedures, and equipment availability for injection turnaround within one day, in any normally expected weather conditions;
OE10		00053	Provisions for reverse well flow of CO_2 and brine, reverse pipeline flow and stack venting of CO_2, and brine separation and disposal;
OE10		00054	Plan, procedures, and equipment availability for immediately shutting in the injection well and suspending CO_2 injection for an indefinite period;
OE10		00055	Presence and continuous functionality of a compression facility shutoff mechanism that responds automatically (plus manually) to significant changes in pressure or flow rate, and that protects compressors from sudden loss of pipeline pressure;
OE10		00056	Equipment availability and procedures necessary to stop injection on short notice;
OE10		00057	Preliminary plan for a Start-up-Shut-in On Paper (SUSIOP) exercise comparable to existing DWOP and CWOP exercises.
OE11	EQUIPMENT SPARING PLAN	00074	For each project mechanical system (including all project equipment assets): Develop a list of critical equipment, equipment available for acquisition, maintenance schedules, and replacement lead times (procurement plus installation).
OE11		00075	Ensure that both operational and monitoring equipment are included in equipment sparing plan.
OE11		00076	Estimate the loss severity/hazard/risk associated with loss/malfunction of each item of critical equipment, and establish a risk-based, item-specific sparing policy.
OE11		00077	Identify suppliers of backup equipment.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
OE12	OPERATIONAL MONITORING PLAN	00078	To enable quick response to protect HSE in the event of a hazardous condition, develop a plan to continuously monitor live data streams from sensors measuring operational and key surface environmental parameters. Ensure the plan addresses at least these elements:
OE12		00079	Automated vs. staffed monitoring;
OE12		00080	Visual inspections;
OE12		00081	Implications of anticipated false-positive and false-negative rates;
OE12		00082	Predefinition of sensor readings that trigger shutdown of any part of injection system;
OE12		00083	Information flow and communications issues;
OE12		00084	Both the physical and the stakeholder (human relations and communications) impacts of notification and shutdown scenarios;
OE12		00085	Lessons Learned from a SUSIOP exercise, if carried out;
OE12		01006	Maintenance of system safety (preventing hazardous conditions and damaging impacts) in case of malfunction of electronic or computer-controlled systems.
PM01	OVERALL RISK MANAGEMENT	00086	Develop and execute a risk management process that includes at least these elements:
PM01		00087	Risk identification;
PM01		00088	Risk evaluation;,
PM01		00089	Actions to minimize risks;
PM01		00090	Documentation and communications that provide assurance to stakeholders;
PM01		00091	Documentation adequate to demonstrate that engineering and geological best practices have been followed. (Note: Rewording from prior was legal proceedings.);
PM01		00092	An IT risk-management solution similar to or an adaptation of the Schlumberger Quest system;
PM01		00240	Consider moving facilities and injection-related tasks in OE01 and OE04 to other RRAGs for which R person is on site (ISGS, Carbon Services IL team, ADM);
PM01		00241	Identify which RRAs are critical to complete before injection start;
PM01		00242	Documentation and communications that provide assurance to stakeholders about risk evaluation and management.
PM02	PERMITS	00093	Write permitting and compliance plan, including the following specific details:
PM02		00094	Complete listing of needed permits and potential permit modifications;
PM02		00095	Listing of permits potentially needed at short notice (such as for workovers) and their Information requirements;
PM02		00096	Permit-issuance schedule/forecast;

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
PM02		00097	Resource estimate for permitting-agency communications and support;
PM02		00098	Tables and schedules documenting all permit requirements and their assignment for execution (RACI);
PM02		00099	Archiving/recordkeeping plan for agency communications regarding permits;
PM02		00100	Development of completion report plan and assignment for execution (RACI).
PM03	DATA MANAGEMENT PLAN	00101	Establish written protocols for management of all classes of data collected by the project, including: continuous/episodic/irregular data; numerical/textual/image/audio data; data originating in loggers/laptops/mainframes/cameras/handwritten, etc.
PM03		00102	Include procedures to prevent premature distribution of raw and/or un-reviewed data.
PM03		00103	Ensure plan includes guidelines and policy for data collection and upload; technical (QA/QC) and administrative review; storage, archiving, general and limited distribution, and MGSC partner access.
PM03		00104	Ensure data is kept accessible to project partners, as needed, and as consistent with policy.
PM03		00105	Establish criteria for public data release and publishing, consistent with Communications Plan.
PM03		00106	Consider adapting existing Information Technology and Schlumberger Integrated Project Management solutions such InterACT* global connectivity, collaboration, and information service.
PM03		00107	Maintain data in the simplest common format in which data integrity can be preserved, e.g., character delimited format, platform independent, on robust storage media.
PM03		01001	Maintain records of project activities to support defense against not-unlikely claims of damage, in conjunction with baseline environmental measurements.
PM04	BUDGET	00108	Ensure that budgeting plan addresses at least these elements:
PM04		00109	Ongoing awareness of funding status with funding agency;
PM04		00110	Ongoing efforts to reduce cost- and budget-sourced risks;
PM04		00111	Mission Sensitivity assessment for potential budget changes in each budget area;
PM04		00112	Cost estimates (to partnership and/or partners) for potential events with significant cost implications such as groundwater contamination;
PM04		00113	Review of possible insurance coverages and costs;
PM04		00114	Cost-control strategies in subcontractor relationships;
PM04		00115	Quarterly and event-driven Approval for Expenditure review;
PM04		00116	Review of potential cost-share arrangements with other Regional Carbon Sequestration Partnerships;

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
PM04		00117	Cost-based review of Sparing and Procurement plans;
PM04		00118	Contingency planning for unusual monitoring, emergency repairs, or other purposes;
PM04		00119	Contingency planning for general budget shortfalls.
PM05	STAFF AND COORDINATION	00120	Ensure that Project Management Plan addresses at least these elements:
PM05		00121	Project overview, organization, and structure;
PM05		00122	Risk management approach;
PM05		00123	Project schedule, milestones, and deliverables;
PM05		00124	Timing flexibility/criticality;
PM05		00125	Budget summary and cost plan;
PM05		00126	Success criteria and decision points;
PM05		00127	Regular staff communications;
PM05		00128	Regular coordination among technical and task leaders;
PM05		00129	Staff continuity / expertise maintenance;
PM05		00130	Verification of critical-task completion.
PM05		00131	Organize the formation of self-run, self-led technical teams to review data and plans and to make timely technical recommendations for certain project activities for which concentrated and diverse technical expertise is needed to support decision making – especially where there may be competing objectives.
PM05		00132	Ensure that teams are organized around at least the following activities: seismic survey design, core-point definition and core disposition, unanticipated events, and injectivity backup and goal triage (see separate RRAG's).
PM08	COMMUNICATIONS POLICY	00142	Ensure that a project communications policy addresses at least these elements:
PM08		00143	Release of raw data and modeling results;
PM08		00144	Communications about possible negative events;
PM08		00145	Linkage of project with particular societal goals and/or industries;
PM08		00146	Internal publishing and execution of the communications policy itself;
PM08		00147	Staff training;
PM08		00148	Review processes.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
PM12	INJECTIVITY BACKUP PLAN	00178	If characterization and/or operations data appear to show less injectivity (either expected or actual) than needed, organize project group to evaluate options including: data reacquisition/verification; additional injection points (in same or different well); controlled fracturing; proceeding with lower than desired injectivity, and others.
PM13	CO_2 MOVEMENT CONTINGENCY PLAN	00179	Write contingency plans for unexpected CO_2 movement that address at least the following scenarios (x):
PM13		00180	(x) Data showing CO_2 laterally migrated significantly beyond expectations based on models; and
PM13		00181	(x) CO_2 present in non-design locations in injection-well casing or annulus.
PM13		00182	Address public communications for these events in this plan or the general Communications Plan.
PM13		01002	(x) Significant atmospheric CO_2 release (significant in this context means a release large enough that mechanical means of ensuring rapid dispersal might be applied, if available).
PM13		01003	(x) Discovery of a monitoring well or non-project well providing a leakage pathway for CO_2 or brine;
PM13		01004	(x) Data showing CO_2 above the Eau Claire caprock;
PM13		01005	(x) Pipeline break, emphasizing protection of injection wellhead and completion;
PM13		01007	(x) Abrupt wellhead rupture/shear below master valve, caused by accident or intention;
PM14	GOAL TRIAGE	00183	Ensure that specific individuals are tasked to monitor progress toward research goals and to communicate to project managers should goal attainment be in doubt.
PM14		00184	Develop plan for convening project leaders, partner representatives, and NETL representatives to prioritize goals and potential responses in the event that goal attainment appears significantly threatened.
PM15	COMMUNICATIONS PLAN	00149	Develop a Communications Plan to establish preparedness for potential Communications challenges, and to timely and accurately convey project info to target audiences.
PM15		00150	Communications Plan element: Text statement of overall purpose and strategy of the Communications Plan.
PM15		00151	Communications Plan element: Descriptions of MGSC consortium, IBDP project, and target audiences.
PM15		00152	Communications Plan element: Guidance text for Listening, Information Gathering, and Influencing; Site Visits; Publications; Crisis Communications.
PM15		00153	Communications Plan element: Prepare policy documents addressing general safety, data distribution, photography, and publications review process.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
PM15	COMMUNICATIONS PLAN	00154	Communications Plan element: Prepare and maintain documents that list project Talking Points and Answers to Frequently Asked Questions.
PM15		00155	Communications Plan element: Plan for frequent monitoring of and content supply to mainstream media and social electronic information media.
PM15		00156	Communications Plan element: Maintain a current, publishable information set on project activities, project status, characterization data, and monitoring data.
PM15		00157	Communications Plan element: Create a publications record that shows project progress and progressively improved subsurface and plume knowledge.
PM15		00158	Communications Plan element: Documentation of and communications about project awareness of, development of, and adherence to industry best practices.
PM15		00159	Communications Plan element: Plan for timely and thorough responses to challenges (in permitting or other contexts).
PM15		00160	Communications Plan element: Plan for project-partner participation in project meetings, conferences, Communications-team briefings, DOE Outreach Working Group, and industry-group meetings.
PM15		00161	Communications Plan element: Description of groundwater usage within AOR; strategy and plans for relations with groundwater stakeholder organizations; publication of project's groundwater protection efforts.
SR01	SITE SELECTION AND AOR	00185	Gather regional information pertinent to site selection and review, including many of the following, depending on associated risks:
SR01		00186	Hydrochemically and geochemically characterize USDWs;
SR01		00187	Ensure that lithostratigraphic data support that effective primary and secondary caprocks extend well beyond AOR;
SR01		00188	Identify and ensure that permitting records and anecdotal drilling history support negligible risk of unknown abandoned wellbores;
SR01		00189	Physically inspect all known abandoned wellbores (or their sites/locations) that reach the aquifer overlying the primary caprock;
SR01		00190	Chemically characterize geologic media through which CO_2-induced movement may occur;
SR01		00191	Characterize historical environmental conditions that might influence the interpretation of monitoring data on the site;
SR01		00192	Establish a well-sampled baseline of environmental monitoring data (and continue to monitor baseline parameters in case of project delay);
SR01		00193	Review groundwater records for potential changes induced near natural-gas storage facilities.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
SR02	USDW BASELINE CHEMISTRY	00194	Establish baseline geochemical and hydrochemical attributes and flow-related characteristics of USDWs within AOR.
SR02		00195	Ensure USDW characterization is adequate to support a later demonstration (if justified by data) that groundwater resources have not been negatively impacted by project.
SR02		00196	Establish baseline (including variation ranges) pH, and concentrations of constituents as specified in applicable permit conditions or of potential concern.
SR03	RESERVOIR CHARACTERIZATION PLAN: IN-ZONE WELLS	00197	Write a detailed plan for data acquisition in wells that penetrate or approach the target injection zone, that delivers required information to the Well Bid Package and Well AFE, and addresses at least the following elements:
SR03		00198	Satisfaction of risk management, regulatory, operational, and research goals;
SR03		00199	Data-acquisition implications for well design and instrumentation;
SR03		00200	Open-hole and cased-hole logging;
SR03		00201	Cuttings samples, and whole and sidewall cores;
SR03		00202	Fluid sampling, including drillstem testing;
SR03		00203	Coordination of objectives with the overall MVA plan;
SR03		00204	Data delivery to the Data Management System.
SR06	MVA PLAN	00027	Implement an observations/data collection plan to be used during a potential leakage event.
SR06		00222	Write an MVA plan that covers all methods and instrumentation used to monitor effects of CO_2 injection and plume migration.
SR06		00223	Address pre-deployment testing, deployment procedures, procedures and frequencies for data collection and sampling, data delivery to Data Management System, and sparing policy for long-lead-time instruments.
SR06		00224	Review planned MVA techniques to ensure they will provide early detection of near-wellbore CO_2 movement out of the reservoir or vertically up the borehole.
SR06		00225	Determine if monitoring data gaps exist that affect the ability to meet data needs for the purposes of HSE protection, plume simulation, and research.
SR06		00226	After deployment of field instruments, verify data collection integrity soon and frequently.
SR06		00227	For data automatically transmitted from remote locations, only expand length of time between data transmissions (or field visits) once data collection and transmission integrity have been demonstrated.
SR06		00228	Periodically review data and plume-simulation forecasts to enhance/adjust monitoring strategy.

RRAG ID	Risk Response Action Group	Treatment ID	Risk Treatment
SR06		00229	If observations or simulations indicate plume migration into area where monitoring is or will be impaired, formulate revised operational and/or surface-access plans.
SR06		00230	Ensure that elements of the MVA program (e.g., chemical sampling and micro- and passive seismic monitoring) and simulation program are adequate to demonstrate deployment of best practices (engineering/geological/geophysical).
SR06		00231	As part of overall MVA plan, assess impact of seasonal, weather, and land-use variability (e.g., cropping, development) on data collection efforts, monitoring equipment, and operational infrastructure.
SR06		00232	Ensure high data quality by selecting and verifying monitoring equipment, methods, and schedule.
SR08	SURFACE SEISMIC	00005	Confirm that seismic operator (WesternGeco) has developed and adheres to an adequate QHSE plan.
SR08		00233	Conduct repeated 2D- and/or 3D-seismic monitoring to track the CO_2 plume, provide information for simulations/modeling, and identify faults and fracture zones.
SR08		00234	Before each field acquisition, conduct seismic plan on paper (SPOP) exercise to anticipate problems and QHSE issues and to optimize data quality.
SR09	STATIC EARTH MODEL	00235	Develop plan for constructing and refining a computer-hosted static earth model, including:
SR09		00236	Model selection and design, data sources, specifications for review and incorporation of new measured data, magnitude and significance of geological uncertainties in terms of plume-migration influence and project risk, and coordination with the Data Management System for geologic characterization data.
SR10	SIMULATION PLAN	00237	Develop plan for performing plume simulations and comparing their results.
SR10		00238	Include specifications for review and incorporation of measured data, model selection and design, nature of output, sensitivity analysis, documentation, and coordination with the Data Management System for geologic characterization data.
SR10		00239	Ensure documentation addresses the results range, confidence in plume prediction/ chance of surprises, and criticality of potential additional measured data from existing or new monitoring locations, and includes information to evaluate and compare different simulations.

7.5 RACI Chart – Stage 1

Organization Abbreviations (C and I roles in RACI chart)	
(c)	Covered by R-A-C positions
all	All partner organizations
A	ADM
D	Drilling contractor
G	WesternGeco
I	ISGS
L	Laboratory partners
M	MGSC management
N	NETL / DOE
R	Regulatory authority
S	Schlumberger Carbon Services
T	Trimeric
V	Vendors
W	Westbay organization

Risk Response Action Group – Topics and RACI Assignments

RRAG Sq #	RRAG ID	Risk Response Action Group	Responsible	Accountable	Consult	Inform
1	OE01	QHSE PLAN: Subsurface and Seismic Operations	Well Designer	Project Manager	IGAD	(c)
2	OE02	QHSE PLAN: Surface and Injection Operations	Site Operations Manager	Principal Investigator	TSI	(c)
3	OE03	SITE SECURITY PLAN	Site Facilities Manager	Site Operations Manager	I	(c)
4	OE04	DRILLING PLANS: DEEP WELLS	Well Designer	Project Manager	ISA	RM
5	OE05	COMPLETION PLAN: IN-ZONE WELLS	Completions Designer	Project Manager	IS	A
6	OE06	COMPLETION PLAN: GEOPHONE WELLS	Assistant Project Manager	Project Manager	IS	A
7	OE07	COMPLETION PLAN: SHALLOW GW MON WELLS	Monitoring Manager	Senior Monitoring Manager	I	A
8	OE08	CO_2-RESISTANT WELL CONSTRUCTION: IN-ZONE WELLS	Completions Designer	Project Manager	SA	I
9	OE09	INJECTION OPERATIONS ENVELOPE	Compression Manager	Principal Investigator	ISA	all
10	OE10	INJECTION OPS AND SHUT-IN PLAN	Compression Manager	Principal Investigator	ISA	all

RRAG Sq #	RRAG ID	Risk Response Action Group	Responsible	Accountable	Consult	Inform
11	OE11	EQUIPMENT SPARING PLAN	Compression Manager	Principal Investigator	AVIWLS	(c)
12	OE12	OPERATIONAL MONITORING PLAN	Compression Manager	Site Operations Manager	SI	M
13	PM01	OVERALL RISK MANAGEMENT	Risk Manager	Project Manager	AIS	IS
14	PM02	PERMITS	Deputy Principal Investigator	Site Operations Manager	STI	I
15	PM03	DATA MANAGEMENT PLAN	Information Technology Manager	Project Manager	SIV	all

RRAG sqno	RRAG ID	Risk Response Action Group	Responsible	Accountable	Consult	Inform
16	PM04	BUDGET	Administrator	Principal Investigator	AST	M
17	PM05	STAFF AND COORDINATION	Deputy Principal Investigator	Principal Investigator	I	M
18	PM08	COMMUNICATIONS POLICY	Deputy Principal Investigator	Principal Investigator	S	all
19	PM12	INJECTIVITY BACKUP PLAN	Reservoir Engineer	Principal Investigator	SA	(c)
20	PM13	CO_2 MOVEMENT CONTINGENCY PLAN	Completions Designer	Principal Investigator	SIATV	all
21	PM14	GOAL TRIAGE	Deputy Principal Investigator	Principal Investigator	S	(c)
22	PM15	COMMUNICATIONS PLAN	Deputy Principal Investigator	Principal Investigator	SA	all
23	SR01	SITE SELECTION AND AOR	Monitoring Manager	Principal Investigator	ASI	RN
24	SR02	USDW BASELINE CHEMISTRY	Monitoring Manager	Senior Monitoring Manager	I	R
25	SR03	RESERVOIR CHARACTERIZATION PLAN: IN-ZONE WELLS	Assistant Project Manager	Reservoir Engineer	SI	RN
26	SR06	MVA PLAN	Monitoring Manager	Principal Investigator	S	A
27	SR08	SURFACE SEISMIC	Principal Geophysicist	Assistant Project Manager	GSAI	(c)
28	SR09	STATIC EARTH MODEL	Principal Geologist	Principal Investigator	SI	(c)
29	SR10	SIMULATION PLAN	Principal Modeler	Reservoir Engineer	IS	(c)

7.6 Stage 2 Scenarios

Scenarios for Stage 2 were topically grouped similarly to Stage 1. Scenarios are sorted from highest to lowest risk within each topic group.

Table: Scenario Groups and Scenarios

Scenario Group	Scenario #	Scenario	Risk Rank in Group
Budget, Cost	S#052	DOE funding is reduced, and not enough funds remain for proper site/project closure.	1
	S#018	After ICCS injection start, IBDP plume does not match predictions. IBDP is required to extend post-injection monitoring period by 10 years, but lacks funding.	2
	S#050	ADM terminates injection and logistical support for IBDP, and many research goals cannot be met.	3
	S#036	After injection is completed, resource reductions prevent analyzing all data being produced and important findings are lost.	4
	S#072	IBDP diverts resources to establish surface seismic monitoring, instead of partnering with another group.	5
	S#074	Manufacturing of key components ceases, project lacks spares, and incurs extra cost for custom manufacture.	6
	S#053	IBDP requests that CCS2 be relocated because of concern about faults, but IBDP must bear the cost.	7
Data Interpretation and Care	S#021	The IBDP project data is required to be made public without time for adequate analysis and/or significant publications from the project team, resulting in misrepresentation of the information.	1
	S#019	As community college becomes more involved, project data become widely available, and a misinterpretation leads to negative public reaction.	2
	S#026	Information stored on external disks becomes corrupted or lost, or critical metadata are lost.	3
	S#017	Emphasis on microseismic monitoring diverts resources and harms performance in geology, reservoir engineering, plume tracking, regulatory reporting, etc.	4
	S#020	Release of project monitoring data (with no technical concerns) unduly causes public concern, resulting in opposition to project.	5
	S#013	DOE requests different time frames for reservoir simulation. Results are unexpectedly affected, showing that plume fate is less predictable than believed.	6
	S#071	Project stops injection and installs intensive new seismic monitoring; then events are shown to be common and harmless, and project is criticized for over-reacting.	7

Scenario group	Scenario #	SCENARIO	Risk Rank in Group
Environmental Monitoring	S#002	Westbay system fails beyond repair and prompts a need for a new Ironton-Galesville monitoring well.	1
	S#004	Sensor breaks or is damaged in accident; delay to replace or repair causes irretrievable information loss.	2
	S#047	Non-project activities (land-surface brine disposal, coal pile runoff, construction) affect environmental compliance data.	3
	S#059	A key monitoring component stops working; not enough injection time remains to establish trends through another method, and closure is delayed.	4
	S#024	Climate or access conditions limit sampling or add variability to environmental monitoring data; regulators lengthen the pre-closure monitoring period.	5
	S#006	A permit-required (or established) monitoring program obstructs meeting a scientific goal (evaluate geochemistry or geomechanics, track plume).	6
Health, Safety, Environment	S#077	Injury from a common industrial or drilling hazard.	1
	S#042	Mechanically caused rupture (e.g., equipment collision) in CO_2 pipeline or wellhead creates toxic atmosphere or IDLH low-oxygen condition.	2
	S#080	After injection is established, focus on HSE declines and an accident occurs.	3
	S#041	Rupture in CO_2 pipeline or wellhead allows rapid gas escape, creating mechanical and temperature hazards.	4
	S#075	Moving vehicle strikes person or damages equipment; traffic hazards onsite or offsite.	5
	S#078	Injury from weather hazards: Cold, heat, wind, lightning, etc.	6
	S#081	Field staff becomes less vigilant; an unsupervised project subcontractor vehicle collides with field equipment.	7
	S#063	Emphasis on safe CO_2 surface operations imposes limitations that impede subsurface monitoring.	8

Scenario group	Scenario #	SCENARIO	Risk Rank in Group
Operations, Mechanical Integrity	S#051	Packer in CCS1 fails and a costly workover is needed.	1
	S#022	Physical process models incorrectly predict effects on engineered components; unexpected equipment failure occurs.	2
	S#045	Critical sensors fail, preventing project from meeting permit requirements and causing shutdown.	3
	S#043	CO_2-acidified brine moves via wellbore, dissolving metals along the way, and into USDW.	4
	S#073	Intermittent or poorly controlled injection causes downhole components to rapidly degrade chemically or mechanically.	5
	S#067	Falloff-test protocol continues to require sharp shut-ins, and a felt seismic event is induced.	6
	S#023	Relations of pressure, temperature, and injection rates are not understood; regulators require shut-ins and testing.	7
	S#076	Management de-emphasizes effort for injection operations, resulting in inadequate reporting and safeguards.	8
	S#049	To avoid induced seismicity, test protocol is changed to step-rate shut-ins, and interpretive value of pressure data is lost.	9
	S#079	Injection is poorly controlled or unsteady, increasing uncertainties in interpreting monitoring data.	10
Plume Footprint	S#001	Plume migrates beneath sensitive area or unexpectedly far, increasing monitoring requirements and cost.	1
	S#084	New and untested technologies malfunction, increasing cost and impairing data acquisition.	2
	S#015	CO_2 plume is confined to a thin interval that cannot be seismically imaged. Closure is not allowed until plume can be mapped through other means.	3
	S#010	Claim of project-caused damage to geologic resource (water, gas, oil, coal), agriculture, or buildings must be legally defended.	4
	S#009	Surface or shallow subsurface condition impairs seismic imaging, groundwater sampling, logging, or other data acquisition, impairing ability to track plume.	5
	S#008	Access denial for surface seismic surveys impairs data quality and prevents the project from tracking plume development.	6
	S#011	Physical process models or simulations incorrectly predict plume fate, causing incorrectly targeted data acquisition.	7
	S#014	After nearby well starts injection, IBDP data can no longer establish single-well plume distribution, harming research value and putting closure at risk.	8
	S#005	CO_2 exits storage reservoir and crosses caprock via fault or fracture.	9
	S#066	CO_2-acidified brine moves via fault, dissolving metals along the way, into USDW.	10
	S#012	Simulations show that ICCS injection may cause the IBDP plume to cross a property boundary after IBDP injection is complete.	11
	S#062	Project surface operations (e.g., seismic data acquisition) impair other land uses, or cause physical damage, resulting in public opposition.	12

Scenario group	Scenario #	SCENARIO	Risk Rank in Group
Public interactions	S#085	An unplanned event occurs; news media become involved; key people are unavailable but a public response is needed.	1
	S#054	After a felt seismic event, public is taken by surprise and demands injection stop.	2
	S#056	Community college and neighbors become vocally opposed to injection, based on aesthetics and concern about earthquakes and hazardous materials.	3
	S#055	Based on independent seismic monitoring, poorly informed conclusions are published that cause public opposition.	4
	S#007	Model results that become public show plume traveling off property controlled by project partner, incurring legal action or payment demand by property owner.	5
	S#057	A small non-project related earthquake is felt in the project area; project microseismicity data become public, and media claim that this proves project caused the felt event.	6
	S#070	Fear of fracking spurs public reaction to microseismic activity; project is ordered to stop injection until it proves safety.	7
	S#040	Public concern about water contamination mobilizes opposition that impedes project activities, arising without a specific triggering event.	8
	S#061	In opposition to fracking or CCS, someone vandalizes seismic-acquisition equipment.	9
	S#069	Prominent environmentalist campaigns against CCS and IBDP based on concern about chemical interaction of CO_2 with subsurface microbes.	10
Regulations, Permitting, Closure	S#046	Regulatory agency is surprised to learn of a connection between seismicity and injection, and requires shutdown pending investigation and additional monitoring.	1
	S#003	CCS2 logs show a fault cutting the Mt. Simon that looks important, and regulators require IBDP injection to stop.	2
	S#035	Change in controlling agency from IEPA to EPA greatly increases administrative burden and makes project goals hard to reach.	3
	S#060	US EPA Class VI permit is denied to IBDP unless many modifications are made, thus ending the project prematurely.	4
	S#064	Project is ordered to stop injection until demonstrating compliance with Class VI rules.	5
	S#058	Permit is modified by agency to require increased public data access; this causes delays, adds work burden, and leads to a failure to meet research objectives.	6
	S#065	Professional community cannot agree on IBDP plume size and recalculated AOR given ICCS influence, so regulators suspend injection permit.	7
	S#016	Gas storage industry petitions regulators to suspend injection while IBDP pressure effects are documented and reviewed.	8

Scenario group	Scenario #	SCENARIO	Risk Rank in Group
Seismicity	S#044	CCS2 increases reservoir pressure, triggers a felt seismic event, and regulators shut down both projects, pending investigation.	1
	S#048	ICCS does not effectively apply IBDP research on injection/shut-in and microseismicity, and induces seismicity that causes regulators to shut down both projects.	2
Staff and Expertise	S#082	A valuable subcontractor has scarce resources and does not send appropriate staff to complete a job.	1
	S#033	Poor scope definition for a monitoring method leads to unreasonable expectations and failure to meet goals.	2
	S#083	Conflicting analyses between two project organizations causes a rift in the relationship, and project work quality suffers.	3
	S#027	An incorrect interpretation of data damages CCS credibility and prompts cancellation of future development.	4
	S#032	A crisis event occurs for which ISGS/Schlumberger Carbon Services are inadequately prepared, and reputation is damaged.	5
	S#030	Staff turnover impairs teamwork and communication; lack of coordination causes poorly informed decision-making.	6
	S#039	Late in project, shift of experienced personnel to new projects damages service quality and final product during closure and documentation phases.	7
	S#068	An operational crisis occurs while key decision-makers are out of country, and situation worsens while project cannot take decisive action.	8
	S#028	A third party evaluates project data and discredits results previously published by the project.	9
	S#034	Project activities, designs, and decisions are sparsely documented, and important CCS operational learning is lost.	10
	S#029	Data processing technique gives results later recognized as wrong, increasing uncertainty about injection effects and the risk of a poor management decision.	11
	S#037	A key person suddenly no longer works for the project, and completion is threatened because objectives, goals, and methods have not been sufficiently shared.	12
	S#031	Staff turnover adds variables in sampling or analysis techniques; adds uncertainty to interpretation.	13
	S#025	The partnering companies' reputations are damaged because the information- sharing opportunities for the project are not realized.	14
	S#038	New staff are unfamiliar with permit requirements; project is burdened with increased regulatory oversight.	15

7.7 Stage 2 Risk Treatments

As for Stage 1 treatments, the correspondence between scenarios and treatments is not 1:1; a single scenario may be addressed by multiple treatments, and vice versa. Generally, treatments are meant to address scenarios in the scenario group of the same name. On a continuing basis, the Responsible and Accountable persons infer the logical connections among scenarios and treatments, and the effectiveness of treatments toward risk reduction. Electronic spreadsheets showing the specific correlations between scenarios and treatments are available.

Trt#	Scenario Group	RISK TREATMENTS
T#048	Budget, Cost	Add Schlumberger Carbon Services' resources to ensure all project duties are properly performed — even when faced with microseismic or unanticipated new concerns.
T#065		Provide adequate time for data analysis.
T#143		Perform geomechanical research to establish a (probabilistic) relationship between operational events and seismicity, and to establish likely bounds to seismic magnitude.
T#145		Work closely with United States Geological Survey (USGS) to establish and conduct project team reviews (in lieu of independent investigations); implement non-disclosure agreements.
T#146		Gather seismic data on plume distribution to image both the IBDP and the adjacent project's plumes — after simultaneous injection has occurred for a sufficient amount of time to generate meaningful data.
T#147		Ensure that adequate funds are budgeted to support project's own surface seismic monitoring without cuts to other monitoring.
T#148		Utilize a team effort approach to allocate responsibilities (assignments) to analyze data and summarize findings (barring unexpected problems near the end of the project).
T#150		Integrate field results into subsurface model to make more accurate simulations and higher probability future realizations of plume locations to support final proposed post-closure monitoring plan.
T#151		Confirm with DOE/ADM that a funding source would be available to support extended monitoring in case of unexpected event or need.
T#049	Data Interpretation and Care	Design microseismic studies, not only to gather site-specific data, but to advance the general understanding of these events and their mechanisms -- so that microseismic monitoring is understood to be a sound research investment even should it lack direct project benefits.
T#050		Ensure that data storage and backup are maintained. Assign a specific person to be responsible for ensuring data are regularly backed up, logged, and archived.
T#051		Continue to improve and update the standards and practices with respect to metadata. Maintain data collection and processing SOPs to ensure metadata integrity.
T#052		Duplicate storage, store backup disks, metadata, and other information in a separate location.
T#053		Provide and require training for Richland Community College personnel handling project data to ensure confidentiality and public sharing protocols and limitations are maintained.

Trt#	Scenario Group	RISK TREATMENTS
T#054	Data Interpretation and Care (cont.)	Facilitate positive public engagement with and through Richland Community College and the National Sequestration Education Center (NSEC), by providing materials and displays that represent our most current and well- founded interpretations.
T#057		Ensure that any Web-based information related to the monitoring data that is intended for public access is actually always readily accessible.
T#058		Contract microseismic monitoring and analysis to a separate group.
T#059		Focus team efforts on the proper interpretation of project data by reducing outside influences.
T#060		Further develop research plans to establish/ensure that project team is proceeding on-pace, on-plan with data evaluations and publication.
T#061		Mitigate or avoid public misinterpretation of project information by having materials prepared to show that the public and the environment are not threatened.
T#063		Place higher priority on data evaluations and publications to facilitate the development of authoritative works.
T#064		Maintain balance in efforts to study injection effects (both hydrodynamic and geomechanical), by convening a technology forum within IBDP to consider best current knowledge about the formation and fluid-flow effects; then develop and investigate hypotheses to explain the observations.
T#068		Publish peer-reviewed works on project data to reduce likelihood of misinterpretation.
T#069		Examine simulation results at regular time intervals throughout the injection period to identify any potential anomalies in plume development over time.
T#070		Organize team effort to allocate responsibilities (assignments) to analyze data and summarize findings.
T#179	Environmental Monitoring	Pull Westbay package and plug right below Ironton-Galesville Formation to facilitate further monitoring.
T#180		Perform in-depth site characterization, research historic land use, and track activity to provide better understanding for potential non-project activities influencing data. Efforts are in place to compile all available land use data.
T#181		Expand some operational costs to ADM (as necessary) to improve site conditions. Environmental sampling would adjust sampling frequency to adapt to extended duration, but the cost should be minimal.
T#182		Continue periodic activities using Westbay system to verify that key components are functioning.
T#183		Reassess project goals periodically, and determine if any are jeopardized, based on permit requirements.
T#184		Collect robust environmental compliance data with corroborating evidence that observed trends are non-project related.

Trt#	Scenario Group	RISK TREATMENTS
T#186	Environmental Monitoring (cont.)	Losing a sensor is likely; delay of replacement is also likely.
T#187		Prepare re-completion plan to remove Westbay and replace with an in-zone and above- zone monitoring system
T#188		Engage with regulators so they are aware of the project ramifications of permit changes. Hold regular reviews to ensure that scientific goals and monitoring program objectives for the project are aligned.
T#189		Identify which sensor is damaged and evaluate existing data to determine if trend using other data can be implemented.
T#190		Design contingent plan to measure Westbay pressure in Ironton-Galesville Formation only.
T#191		Ensure that components are maintained and work properly. Confirm secondary backup processes are in place in the event of Westbay failure; e.g., pressure profiles available to be taken every 2 weeks.
T#192		Continually consult with Westbay managers and technicians to improve reliability of all components of the system. Maintain a collaborative relationship with Westbay. Continue to suggest ways to improve the system.
T#193		Create sparing plan for critical system components, primarily for pressure monitoring, and secondarily for highest-priority fluid sampling locations (e.g., zones 2, 3, 4, 5, 10).
T#194		Limit the reliance on use of zones 10 or 11 to meet permit conditions.
T#195		Limit permit conditions or expectations (where possible) that run counter to project research goals. If goal is highly important, engage with regulators in discussion about any critical need for deviating from permit requirement or established practice.
T#196		Employ periodic groundwater monitoring reports to succinctly explain non-project related nature (if supportable) of significant observed variability of project related monitoring parameters.
T#197		Installing a new well is a relatively easy and viable solution. Permitting and time to drill — including mobilization and demobilization — are issues to be considered; additional challenges come from the potential for lost data and the relocation to a new site.
T#198		Maintain availability of a spare sensor to minimize a potential, costly loss of data.
T#213		Redesign Westbay cable heads to prevent arcing.
T#214		Change tubing fluid to prevent corrosion of Westbay components.

Trt#	Scenario Group	RISK TREATMENTS
T#115	Health, Safety, Environment	Develop HARC for onsite vehicle traffic and equipment movement; employ Journey Management Plan and recurrent training.
T#116		Conduct frequent and regular pre-job safety meetings
T#118		Ensure that project personnel regularly on site are trained and drilled in recognizing and responding to the possible presence of low-oxygen conditions.
T#119		Implement a severe-weather plan.
T#120		In the fall of each year, inventory/order sufficient quantities of insulated Personal Protective Equipment (PPE) for as many staff as are likely to need them during the coming winter.
T#121		Perform a crisis training and drill.
T#122		Perform an audit to ensure adherence to all safety programs; evaluate progress toward safety program goals.
T#123		Facilitate the continuing interest in and focus on the effective culture of safety that has been developed on this project.
T#126		Define specific shut-down conditions.
T#127		Shut down operations during extreme storms.
T#128		Emphasize new HSE training objectives to avoid complacency.
T#130	Operations, Mechanical Integrity	Develop a test protocol that optimizes acquisition of the desired data, using a less-abrupt shut-in that is less likely to induce seismicity.
T#131		Work with regulators to enable testing while avoiding extended shut-ins.
T#132		Engage attorney to develop a preliminary response plan that anticipates information needs that can be partly fulfilled in advance.
T#133		Perform geomechanical research to establish a relationship between pressure increase and leakage probability, given the presence of a fault that is not transmissive under pre-injection conditions.
T#134		Conduct regular RST* reservoir saturation tool runs to track vertical CO_2 movement.
T#135		Develop and implement a sparing plan.
T#136		Review and evaluate all sensors; determine where alternative data could be obtained.
T#137		Develop and maintain a workover cost estimate and adjust, as necessary, for inflation of costs.
T#138		Using 1 ½ years of project experience, evaluate and estimate the point at which the cost of replacing major components supersedes the value of the injection.
T#139		Advertise/provide wide public notice that a routine fall-off test will occur, and there could be a minor felt seismic event.
T#140		Perform geologic research to establish the likely history and characteristics of faults in the project boundaries.
T#141		Regularly sample zones to confirm CO_2 location.

Trt#	Scenario Group	RISK TREATMENTS
T#142		Develop and implement replacement plans already for critical sensors; include sources, contacts, any training needed, etc.
T#215		Slow the pumping rate on annulus pump to reduce wear on packers.
T#216		Implement a valve lubrication program for the wellhead valves and a cable replacement plan for the Westbay winches. Monitor corrosion inhibitor effectiveness. Avoid removing the top cap when making an intervention in CCS1, thereby eliminating the risk of damage to the seal ring in the cap.
T#011	Plume Footprint	Acquire a new 3D vertical seismic profile VSP to address the new reservoir simulation that predicts a thicker plume, to see if the CO_2 plume can be identified.
T#012		Acquire data as early as possible to evaluate any potential need to modify the plan and to determine the potential for accurate imaging.
T#013		Limit plume location uncertainties by frequent updates as the project progresses and more data is collected; based on the updates, the monitoring group must weigh the need for expanding the monitoring footprint.
T#014		Incorporate flexibility into the timing of the surveys.
T#015		Continue current efforts to monitor and document the plume (with an emphasis on record-keeping in the event of future legal action). To minimize the risk of statements being misunderstood, conflicting, or taken out of context, project personnel should be trained and briefed on basic procedures for communicating information to outside parties, and be aware of when to refer the interested parties to officially designated spokespeople.
T#016		Continue public outreach to landowners and tenants who are affected by the seismic operations. Be responsive to public concerns as they arise. Promptly remediate any damages to properties as a result of the seismic operations.
T#017		Continue to be mindful that ground conditions will affect the quality of the seismic imaging, and plan to acquire surveys under similar ground conditions at the same time of year, if at all possible.
T#018		Continue to monitor microseismicity related to the injection.
T#019		Cultivate relationship with local news person skilled at explaining risk concepts. Keep regulators currently informed about microseismic monitoring and activity.
T#020		Extending the survey (depending on size and location) may allow for undershooting and additional data to track, monitor, and understand plume behavior.
T#021		Develop and rank (in order of perceived risk) hypothetical scenarios of damage claims. Ensure that sufficient types and frequencies of data are being collected to determine the impact (or non-impact) to the resource of the highest-risk scenarios. Some lower-severity, but higher-likelihood scenarios may provide opportunities to collect data that demonstrate project stewardship/diligence and could potentially pre-empt or reduce the likelihood of future claims being generated.
T#022		Ensure that the geologic model and resulting simulations are updated as new data become available.

Trt#	Scenario Group	RISK TREATMENTS
T#024		Ensure geologic model and simulations are updated with all of the latest monitoring information; a new uncertainty analysis of the plume development should be completed. Begin planning for monitoring over a larger area and assess budgetary impact.
T#025		Explore and develop the unique research opportunity presented when CCS2 begins injection. Although the IBDP will no longer be able to establish a single-well plume distribution once CCS2 starts to inject, this presents a research opportunity that no other projects have had the opportunity to examine, and is therefore a unique opportunity rather than a risk.
T#026		Identify these conditions and manage expectations of stakeholder (i.e., US EPA/IEPA).
T#027		Prepare materials (including a press pack) for dissemination so project personnel can respond immediately with accurate information if potential project issue is developing.
T#028		Ignore/accept this risk for IBDP; the potential consequences will rest mostly upon ICCS.
T#029		Investigate alternative methods and permit requirements for closure.
T#030		Remain aware of such possibilities developing.
T#031		List the normal bounding conditions and operating ranges (envelope) for critical equipment, then investigate whether (1) Equipment is within envelope under all actual conditions, and (2) Important uncertainty exists about the actual conditions.
T#032		Explore requirements of permitting/regulatory agencies. Early implementation may produce fewer costs; develop contingency plans.
T#033		Maintain pressure monitoring and fluid sampling in the Ironton-Galesville Formation for any potential leakage indicators; maintain groundwater sampling programs
T#034		Maintain pressure monitoring and fluid sampling above the Eau Claire Formation to assess potential changes in the Ironton-Galesville that might indicate leakage.
T#035		Evaluate and apply new technologies as appropriate to project needs.
T#036		Proceed with a geological interpretation review to minimize the chance that faults in the Eau Claire Shale have been missed.
T#037		Perform regular peer reviews to help reduce this risk.
T#038		Run regular RST logs on VW2; check for CO_2 when drilling CCS2. If plume is determined to be spreading too far, then modify perforation scheme in CCS1.
T#039		Run a spinner log to verify that CO_2 is still preferentially going into the bottom set of perforations. If this is still the case, the perforations could be packed off and new perforations could be opened to try to distribute the plume over a larger vertical area.
T#040		Research whether a customized insurance product could support closure costs that extend beyond existing bond(s).
T#041		Establish a rigorous model review committee and determine realistic model scenarios, so that all plausible scenarios are identified. Determine probability that all reasonable reservoir characterization and geologic models will yield results of CO_2 migrating beyond project boundaries.

Trt#	Scenario Group	RISK TREATMENTS
T#042		Establish a rigorous model review committee and determine realistic model scenarios, so that all plausible scenarios are identified. Develop contingency plans for data acquisition to minimize the impact of model inaccuracies due to incorrectly targeted data acquisition.
T#043		There is no treatment; two injection wells will interfere with the IBDP objective to understand the plume development around a single injector. IBDP scope and objective should be formally changed with DOE. Combine projects, share resources and objectives, and continue to monitor single plume development at higher rates, pressures, and ultimate injected volume.
T#044		Assuming this scenario is current and correct, blocking off the current perforation and perforating in the zones above the current injection provides one option to force the plume into a greater thickness of the reservoir.
T#045		Determine if there is a high likelihood of this occurring. If so, then evaluate current seismic studies, review plume image data, and determine if the area of coverage needs to be expanded; if not, then evaluate the need for drilling a new well (or wells) to verify the plume.
T#046		Run VSP at most optimum time to get best results. Multiple runs might be scheduled if budget allows and client has confidence that the plume can be imaged. Sampling at CCS2 might serve as a secondary indicator of CO_2 migration and support modeling improvements.
T#047		Work to improve relations with permittees who provide access to property. Draft follow-up letters thanking the permittees for access and provide basic project update.

Trt#	Scenario Group	RISK TREATMENTS
T#054	Public interactions	Facilitate positive public engagement with and through Richland Community College and NSEC, by providing materials and displays that represent the project's most current and well-founded interpretations.
T#056		Provide frequent updates and points of contact to local news media to promote project-appropriate questions and focus, and to ensure timely reporting of accurate information.
T#067		Provide regularly updated fact sheets about the project to the college staff; stress the importance of adhering to this known information, but stress the need for preparation for responding to alternate interpretations.
T#152		Acquire additional monitoring devices to bolster the project's ability to track and identify acoustic events.
T#153		Update the communications plan to include alternate points of contact to address a situation in which the normally designated media-contact people are unavailable, but media contact must be made regardless.
T#154		Prepare fact sheets summarizing seismic events at other CO_2 injection projects, the energy released, and how felt events compare to damaging events.
T#155		Prepare fact sheets: 1) Summarizing CO_2-related monitoring results from near-surface instruments; and 2) Demonstrating the pure-CO_2 composition of the injected flow stream. List CO_2-containing products commonly used by the public, and reiterate that CO_2 is the only potential product being stored underground at the site.
T#155b		Prepare fact sheets that show the CO_2 content in common products that are safely used by the public.
T#157		Conduct baseline and ongoing ground water sampling, and publish and archive data so that non-impact to groundwater can be demonstrated.
T#158		Create bullet-point information sheets for those not to speak with media so they can give appropriate answers if contacted.
T#160		Create factsheets addressing induced seismicity. Publish data so that speculative interpretations do not impact the project. Follow proactive communication strategy.
T#161		Create project information sheet to distribute.
T#163		Develop outreach contingency plan for a felt seismic event.
T#164		Prepare and publicly distribute fact sheets on seismicity.
T#165		Create a surface monitoring system with a traffic light operating procedure in place for various seismic events tied to injection pressures and/or rates. Continual monitoring may assist in avoiding premature or permanent shutdown.
T#166		Discuss responses — internally and in advance — to felt seismicity and agree on possible project responses to such events.

Trt#	Scenario Group	RISK TREATMENTS
T#167		Perform lab tests with formation brine, CO_2, and cuttings from the faulted formations, tracking the changes in metal concentrations, providing basis for reassurance or concern.
T#168		Maintain secure data storage and analysis to preserve past event activity for reference.
T#169		Create a designated backup plan that documents procedure and assigns responsibility to key personnel.
T#170		Promote internal discussion to develop a general public response for a felt seismic event (either due to injection or an event elsewhere).
T#171		Implement a plan that provides for the development of materials ready to communicate to the public, and defines and trains media points of contact. Potentially include this topic in the crisis exercise.
T#172		Run a crisis simulation.
T#173		Establish a rigorous model review committee and determine realistic model scenarios, so that all plausible scenarios are identified. In all communications, distinguish plausible scenarios from possible, but highly unlikely scenarios.
T#174		Ensure the communications plan is followed, the public is aware of current events at the site, and that project personnel are focused on closely monitoring microseismic events.
T#175		Review press-contact procedures established at beginning of project; ensure that newer staff members are aware of procedures.
T#176		Reduce public concerns through a larger-scale private well sampling effort (possibly with help from State Water Survey) focused on drinking water safety.
T#178		Develop crisis plan; update communications plan, have talking points prepared, understand and be ready to explain project response to seismic activity.
T#199	Regulations, Permitting, Closure	Ensure that dialog with EPA is maintained, and EPA remains satisfied with progress toward Class VI satisfaction, so that such an order is not issued.
T#200		Complete the million-tonne injection program as expeditiously as possible under the terms of the current permit.
T#201		Make progress toward Class VI requirements, and negotiate with EPA while continuing injection and monitoring. Pursue budget increase from DOE based on permit requirements.
T#202		Create a reduction in risk by investigating the likelihood of felt seismicity; diligent learning in the demonstration project contributes to project goals and results in a fundamental success factor.
T#203		Conduct geomechanical research to establish a (probabilistic) relationship between pressure increase and seismicity, and to establish likely bounds to seismic magnitude; engage in dialog with regulators and share information prior to any even occurrence.
T#204		Prepare to modify project goals (within budgetary constraints) if regulatory requirements change substantially.

Trt#	Scenario Group	RISK TREATMENTS
T#205		Hold project technical peer reviews with all stakeholders to prevent ordinary technical disagreements from hindering project progress.
T#206		Reduce potential impact by ensuring that data are stored, and the storage is structured in ways that could permit easy, limited public access if this were required.
T#207		Assemble a communications plan to familiarize regulators with induced seismicity and its effects. Use a literature search to identify other documented, existing induced seismicity.
T#208		Continue to examine the subsurface, using best monitoring techniques available, to investigate whether microseismic events are associated with a fault in the subsurface.
T#209		Perform a more detailed inspection and interpretation of all seismic data (by an interpretation expert) to further project certainty. Gather data from the CCS2 well that can help define the risk.
T#210		Create a surface monitoring system with a traffic light operating procedure in place for various seismic events tied to injection pressures and/or rates. Continual monitoring may assist in avoiding premature or permanent shutdown.
T#211		Ensure that all project data are provided to ADM (as permit holder), so they can communicate fully with IEPA, as required and as prudent, to avoid unexpected reactions from regulators.
T#212		Prepare to respond with model results and various calculations of potential energy releases from estimated fault dimensions using in-situ stresses, pore pressure increases over formation pressures, etc.
T#002	Seismicity	Mitigate the risk of induced seismicity by installing seismographs around the IBDP project site. Onsite monitoring could reduce risk, but there doesn't appear to be a way to completely eliminate the risk.
T#003		Prepare a rigorous modeling exercise to predict if increased pressures will be produced. Integrate the new seismic interpretation with the existing models, then calibrate to monitoring results.
T#004		Employ current project knowledge to reduce the factors that may cause an induced event; reduce all avoidable shut downs and try to reduce the pressure shock to the subsurface.
T#006		Enhance knowledge of what pressure changes initiate microseismic events such as stepped increases or decreases, thus helping to better understand limits.
T#007		Develop a communication plan (authored by project professionals) that pre-determines specific responses to regulators, the appropriate recipients of the information, and the best delivery method(s).
T#009		Maintain close working relationships between ICCS and IBDP teams. Ensure both project teams receive monthly reports; budget adequate funds for continued study of seismic activity and monitoring; evaluate any seismic mitigation strategies to minimize events, including perforation plan for CCS2, gradual shut-down of injection activity (where possible), and seismometer deployment.

Trt#	Scenario Group	RISK TREATMENTS
T#066	Staff and Expertise	Provide incentive and assistance to staff to facilitate publication of data in peer-reviewed articles and appropriate professional venues; plan to develop a best practices manual. Note: Additional resources and a self-motivated staff are likely necessary for completion.
T#075		Encourage all professional staff involved in both ISGS and Schlumberger Carbon Services to respond carefully and thoughtfully and suggest interested parties contact key project communications personnel.
T#076		Avoid being in situation with only one domain expert (e.g., one drilling engineer, one reservoir engineer, etc.).
T#077		Increase support staff; add-on site record keeper and archivists as necessary.
T#078		Ensure, as standard practice, that work products and new workflows are internally reviewed before results are applied.
T#079		Conduct periodic permit requirement refresher sessions so each staff member is reminded of relevant permit requirements. Assemble a quick reference sheet of permit requirements.
T#080		Continue to conduct and present world-class scientific research. Improve communications internally in ISGS, and between ISGS and Schlumberger Carbon Services.
T#082		Review data management procedures to ensure no learnings or raw data are lost.
T#083		Designate spokesperson(s) in advance for various scenarios.
T#084		Develop a strategy to retain key staff and provide for hand-off of key responsibilities in events when turnover is anticipated.
T#086		Ensure that workflows for all sampling and analysis procedures are written in detail; always record which staff member performed the procedure, and when possible, employ dual sampling or analysis to ensure standardization.
T#087		Establish a Technical Review board to review all interpretations. Board should consist of an independent group of consultants with oil and gas experience, but not include Schlumberger Carbon Services or ISGS staff.
T#088		Establish concurrence amongst participants about clear expectations and responsibilities.
T#089		Establish SOPs, project guidelines, project procedures, project responses, etc. so that key expectations, goals, procedures, activities, and responses are well understood by all pertinent project personnel.
T#090		Evaluate scope definitions more frequently for key monitoring methods with all relevant stakeholders/participants.
T#091		Establish a project center with ADM, ISGS, and Schlumberger Carbon Services at the same office location, or conduct weekly in-person sessions.
T#092		Identify key personnel. Based on current knowledge, assess their likelihood of departure from the project.

Trt#	Scenario Group	RISK TREATMENTS
T#093		Implement objective of regular documentation and distribution of work (i.e., Quarterly status update from each employee).
T#094		Establish team reviews of calculations, processing techniques, and interpretations.
T#095		Involve internal and selected external peer-review to vet summaries and research.
T#096		Maintain a system of review by people outside the daily operations (peer review).
T#097		Maintain trained backup work force to be available on call. Obtain training for all associated personnel to provide support for well-site operations.
T#098		Ensure that no new activity is started without a clear HARC and JSA being performed. If resources are inadequate, then the well intervention must not be undertaken
T#099		Periodically review expectations for in-place monitoring systems, based on data gathered to date; forecast adequacy of data for eventual project closure.
T#100		Share the established operational contingency plans widely within the group.
T#101		Plan ahead; prepare for the unexpected by having competent local contractors available to assist in operations.
T#102		Post call-out sheet of staff, contractor, emergency response, and support-service contact information for various situations (e.g., mechanical failure, leakage, injury/casualty).
T#103		Prepare crisis simulation plan with professionals in this field.
T#104		Mitigate surprise; ensure that regulatory agency is kept well and accurately informed.
T#105		Reassess project goals with partners and technical staff on a regular basis. Review goals monthly with Schlumberger Carbon Services stake holders. Meet at least bi-weekly with project partners.
T#106		Hold regular meetings between ICCS and IBDP to improve collaboration and information sharing, and to avoid conflicts.
T#107		Review and renew emergency response and contingency plans.
T#108		Provide occasional reminders of press contact procedures and limitations (per Communications Policy).
T#109		Run a crisis simulation drill.
T#110		Ensure that papers from the project undergo thorough reviews from independent technical group prior to publishing.
T#111		Thoroughly prepare information and interpretations. Provide adequate time for data reduction, exploration, and interpretation.
T#112		Employ tools like SharePoint and other human resources tools to train replacement staff.
T#113		Verify goals are attainable by periodic reassessment of progress or ability to meet goals — based on scope definitions and established responsibilities and expectations.

7.8 ROCI Chart – Stage 2

	Scenario Group / Risk Treatment Group	Responsible	Oversight	Consult	Inform
1	Plume Footprint	Principal Modeler	Subsurface Lead	I	ARN
2	Data Interpretation and Care	Principal Geophysicist	Subsurface Lead	I	all
3	Staff and Expertise	Deputy Principal Investigator	Principal Investigator	S	A
4	Health, Safety, Environment	Assistant Project Manager	Deputy Principal Investigator	A	MR
5	Operations, Mechanical Integrity	Assistant Project Manager	Site Operations Manager	I	(c)
6	Budget, Cost	Project Manager and Administrator	Deputy Principal Investigator	M	all
7	Seismicity	Assistant Project Manager	Principal Geophysicist	L	MN
8	Public interactions	Communications Coordinator	Deputy Principal Investigator	A	all
9	Environmental Monitoring	Monitoring Manager	Senior Monitoring Manager	L	R
10	Regulations, Permitting, Closure	Assistant Project Manager	Project Manager	IA	MRN

Organization Abbreviations (C and I roles in ROCI chart)	
(c)	Covered by R-O-C positions
all	All partner organizations
A	ADM
I	ISGS
L	Laboratory partners
M	MGSC management
N	NETL / DOE
R	Regulatory authority
S	Schlumberger Carbon Services

7.9 Related Publications

Technical Papers

- Ken Hnottavange-Telleen, Ivan Krapac, Claudia Vivalda, *Illinois Basin – Decatur Project: initial risk-assessment results and framework for evaluating site performance*, Energy Procedia, Volume 1, Issue 1, February 2009, Pages 2431-2438, ISSN 1876-6102, http://dx.doi.org/10.1016/j.egypro.2009.02.004
  http://www.sciencedirect.com/science/article/pii/S187661020900647X

- Ken Hnottavange-Telleen, Ethan Chabora, Robert J. Finley, Sallie E. Greenberg, Scott Marsteller, *Risk management in a large-scale CO_2 geosequestration pilot project, Illinois, USA*, Energy Procedia, Volume 4, 2011, Pages 4044-4051, ISSN 1876-6102, http://dx.doi.org/10.1016/j.egypro.2011.02.346
  http://www.sciencedirect.com/science/article/pii/S1876610211006254

- Ken Hnottavange-Telleen, Risk Management Status, *Illinois Basin Decatur Project - internal report to IBDP partners*, July 2012

- Ken Hnottavange-Telleen, *Common Themes in Risk Evaluation Among Eight Geosequestration Projects*, Energy Procedia, Volume 37, 2013, Pages 2794-2801, ISSN 1876-6102, http://dx.doi.org/10.1016/j.egypro.2013.06.164
  http://www.sciencedirect.com/science/article/pii/S1876610213004074

- Ken Hnottavange-Telleen, *Risk management at the Illinois Basin – Decatur Project: A FEPs-based approach*, Greenhouse Gases: Science and Technology Special Issue: Illinois Basin - Decatur Project, Volume 4, Issue 5, pages 604-616, October 2014.
  http://onlinelibrary.wiley.com/doi/10.1002/ghg.1450/abstract

Presentations and Posters

- Ken Hnottavange-Telleen, *IBDP Initial Risk Assessment Results and Framework for Evaluating Site Performance*, 7th Annual Conference on Carbon Capture and Sequestration, Pittsburgh, PA, USA (poster).
- Ken Hnottavange-Telleen, *Slicing up the Pie Chart of Risk: Risk Analysis Methods and Early Results from MGSC Phase III*, 7th Annual Conference on Carbon Capture and Sequestration, Pittsburgh, PA, USA.
- Ken Hnottavange-Telleen, Robert Finley, Sallie Greenberg, Scott Marsteller, Ivan Krapac, Jim Kirksey, Randy Locke, *Second Iteration: Re-Assessing CCS Project Risk Using New Data*, AAPG/SEG/SPE Hedberg Conference, Vancouver, BC, Canada, August 16-19, 2009 (poster).
  http://www.searchanddiscovery.com/abstracts/pdf/2010/hedberg_vancouver/#H
- Ken Hnottavange-Telleen, *How Risk Management Guides Decision-Making in Two Large-Scale CCS Pilots*, 8th Annual Conference on Carbon Capture and Sequestration, Pittsburgh, PA, USA.
- Ken Hnottavange-Telleen, *Risk Assessment, Parameter Uncertainties, and Data Acquisition*, IEAGHG Risk Assessment Network meeting #4, Melbourne, Australia, April 16-18, 2009.
- Ken Hnottavange-Telleen, *Progress with Information Capture for CCS Risk Assessment*, IEAGHG Risk Assessment Network meeting #5, Golden, CO, USA, May 17-19, 2010.